NILA SOCIAL MEDIA

A MINI PROJECT REPORT

Submitted by

VAISHNAVI GIREESH

MLM25MCA-20257

To

the APJ Abdul Kalam Technological University in
Partial fulfillment of the requirements for the
award of the degree

of

MASTER OF COMPUTER APPLICATIONS

Under the Guidance of

Ms. ASHWANI VIJAYACHANDRAN

(Assistant Professor, Department of Computer Applications)

DEPARTMENT OF COMPUTER APPLICATIONS

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

(Affiliated to APJ Abdul Kalam Technological University)

OCTOBER 2026

MANGALAM COLLEGE OF ENGINEERING

Accredited by NAAC& ISO 9001:2000 Certified Institution

DEPARTMENT OF COMPUTER APPLICATIONS

VISION

To become a centre of excellence in computer applications,competent in the global ecosystem with technical knowledge,innovation with a sense of social commitment.

MISSION

- To serve with state of the art education,foster advanced research and cultivate innovation in the field of computer applications.
- To prepare learners with knowledge skills and critical thinking to excel in the technological landscape and contribute positively to society.

Program Educational Objectives

- PEO I :Graduates will possess a solid foundation and in-depth understanding of computer applications and will be equipped to analyze real-world problems, design and create innovative solutions, and effectively manage and maintain these solutions in their professional careers.
- PEO II: Graduates will acquire technological advancements through continued education, lifelong learning and research, thereby making meaningful contributions to the field of computing.
- PEO III: Graduates will cultivate team spirit, leadership, communication skills, ethics, and social values, enabling them to apply their understanding of the societal impacts of computer applications effectively.

Program Specific Outcomes

- **PSO I**: Apply advanced technologies through innovations to enhance the efficiency of design and development.
- **PSO II**: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

MAPPING OF PO-PSO-SDG

1. MAPPING WITH PROGRAM OUTCOMES (POs):

SL.NO	POs ADDRESSED	RELEVANCE TO PROJECT
1	PO1	Applies programming, database and web-development concepts.
2	PO2	Addresses social networking, privacy and content-management problems.
3	PO3	Develops a modular web-based social media platform.
4	PO5	Uses Python, Flask, SQLAlchemy, SQLite and AI-related tools
5	PO8	Implements privacy, security and responsible social-media features.

LIST OF PROGRAM OUTCOMES (POs):

PO1 (**Foundation Knowledge**): Apply knowledge of programming logic, database concepts and coding fundamentals for solution architecture and problem solving.

PO2 (**Problem Analysis**): Identify, review, formulate and analyze problems in traditional complaint reporting with focus on user requirements.

PO3 (**Development of Solutions**): Design and develop an innovative online complaint reporting solution for efficient complaint management and tracking.

PO4 (**Modern Tool Usage**): Select and apply modern computational tools and technologies such as PHP, MySQL, JavaScript and NLP for system development.

PO5 (**Individual and Teamwork**): Function and communicate effectively as an individual or team member during the planning, development and testing of the project.

PO6 (**Project Management and Finance**): Apply project management principles such as planning, scheduling, resource management and work breakdown structure.

PO7 (**Ethics**): Follow professional ethics and cybersecurity practices to protect user information, complaint data and digital evidence from unauthorized access.

PO8 (Life-long learning): Develop change management skills and continuously learn and adapt to contemporary technologies and methods for improving the complaint reporting system.

2. MAPPING WITH PROGRAM SPECIFIC OUTCOMES (PSOs):

LIST OF PROGRAM SPECIFIC OUTCOMES (PSOs):

PSO 1: Apply advanced technologies through innovations to enhance the efficiency of design development.

SL.NO	PSOs ADDRESSED	RELEVANCE TO PROJECT
1	PSO1 – Apply advanced technologies through innovations to enhance the efficiency of design development	Advanced technologies such **as** Flask, AI-assisted features, JavaScript and database technologies were applied to design and develop an efficient social media platform.
2	PSO2 – Apply the principles of computing to analyze, design and implement sustainable solutions for real-world challenges	Computing principles including data modelling, database management, web development, security and software design were applied to build a practical social networking solution.

PSO 2: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

3. MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs):

SDG NO	SDGs ADDRESSED	RELEVANCE TO PROJECT
SDG 4	Quality Education	The platform can support knowledge sharing by allowing users to create, share, and access educational and informative content.
SDG 9	Industry, Innovation and Infrastructure	The project promotes digital innovation by integrating AI/ML technologies such as personalized recommendations, automatic caption/hashtag generation, and image analysis into a social media platform.
SDG 10	Reduced Inequalities	The platform provides an inclusive digital space where users from different backgrounds can participate, communicate, and share content.

SUSTAINABLE DEVELOPMENT GOALS (SDGs):

SDG 1 – No Poverty: End poverty in all its forms everywhere.

SDG 2 – Zero Hunger: End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

SDG 3 – Good Health and Well-Being: Ensure healthy lives and promote well-being for all at all ages.

SDG 4 – Quality Education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

SDG 5 – Gender Equality: Achieve gender equality and empower all women and girls.

SDG 6 – Clean Water and Sanitation: Ensure availability and sustainable management of water and sanitation for all.

SDG 7 – Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable and modern energy for all.

SDG 8 – Decent Work and Economic Growth: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

SDG 9 – Industry, Innovation and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialisation and foster innovation.

SDG 10 – Reduced Inequalities: Reduce inequality within and among countries.

SDG 11 – Sustainable Cities and Communities: Make cities and human settlements inclusive, safe, resilient and sustainable.

SDG 12 – Responsible Consumption and Production: Ensure sustainable consumption and production patterns.

SDG 13 – Climate Action: Take urgent action to combat climate change and its impacts.

SDG 14 – Life Below Water: Conserve and sustainably use the oceans, seas and marine resources.

SDG 15 – Life on Land: Protect, restore and promote sustainable use of terrestrial ecosystems, manage forests sustainably, combat desertification and halt biodiversity loss.

SDG 16 – Peace, Justice and Strong Institutions: Promote peaceful and inclusive societies, provide access to justice for all and build effective, accountable and inclusive institutions.

SDG 17 – Partnerships for the Goals: Strengthen the means of implementation and revitalise the global partnership for sustainable development.

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

DEPARTMENT OF COMPUTER APPLICATIONS

OCTOBER 2026



DECLARATION

I hereby certify that the work presented in the project entitled "NILA SOCIAL MEDIA" submitted to the DEPARTMENT OF COMPUTER APPLICATIONS is an authentic record of my own work carried out under the supervision of Ms. Ashwani Vijayachandran, Assistant Professor. This work has not been submitted to any other institution or university for the award of any other degree. The report has been checked for plagiarism by the college and the similarity index is within the permissible limit set by the college.

Date:

Place:

Name & Signature of Student

VAISHNAVI GIREESH

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

DEPARTMENT OF COMPUTER APPLICATIONS

OCTOBER 2026



CERTIFICATE

*Certified that the report entitled "**Nila Social Media**" submitted by **VAISHNAVI GIREESH (MLM25MCA-2057)** to the **APJ Abdul Kalam Technological University** in partial fulfillment of the requirements for the award of **the Degree of Master of Computer Applications** is a bonafide record of the MINI PROJECT work carried out by the student under our guidance and supervision during the academic year 2025-27.*

PROJECT GUIDE
Ms. Ashwani Vijayachandran
Assistant Professor
Department of Computer Applications

PROJECT COORDINATOR
Ms. Athira R Kurup
Assistant Professor
Department of Computer Applications

HEAD OF THE DEPARTMENT
Ms. Divya S B
Associate Professor
Department of Computer Applications

INTERNAL EXAMINER

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I am greatly indebted to the authorities of Mangalam College of Engineering for providing the necessary facilities to successfully complete my Project on the topic **"*Nila Social Media*".**

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VAISHNAVI GIREESH

MLM25MCA-2057

ABSTRACT

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NILA SOCIAL MEDIA is a web-based social networking application designed to provide users with an integrated platform for communication, content sharing, and social interaction. The system allows users to create accounts, manage personal profiles, share multimedia content, and interact with other users through likes, comments, follows, stories, reels, and messaging. It also provides features such as notifications, search and exploration, privacy controls, and account management to support a complete social networking experience.

The application is designed with a modular architecture that separates user management, content handling, social interactions, communication, and administrative functions. Authentication and access-control mechanisms are incorporated to protect user accounts and provide appropriate access to system features. Media upload and management functions allow users to share images and other supported content through the platform.

An additional focus of NILA SOCIAL MEDIA is the integration of AI-assisted functionality. These features are intended to support selected tasks such as content assistance, recommendation-related processing, and image or content analysis, depending on the implemented module. The use of AI provides an opportunity to enhance user interaction while keeping the application suitable for local development and demonstration.

The project demonstrates how a modern social networking application can combine conventional web technologies with AI-assisted features within a single system. The modular design also provides a foundation for future improvements, including more advanced AI capabilities, scalability, enhanced recommendation mechanisms, and additional communication and multimedia features.

Mapping with Sustainable Development Goals (Mention the Goal)	SDG 4 Quality Education SDG 9 Industry, Innovation and Infrastructure SDG 10 Reduced Inequalities

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LIST OF ABBREVIATIONS

ABBREVIATION		FULL FORM
AI	-	Artificial Intelligence
API	-	Application Programming Interface
BLIP	-	Bootstrapping Language-Image Pre-training
CLIP		Contrastive Language–Image Pre-training
CPU	-	Central Processing Unit
CRUD	-	Create, Read, Update, Delete
CSS	-	Cascading Style Sheets
DB	-	Database
DFD	-	Data Flow Diagram
ER	-	Entity-Relationship
GET		GET Request Method
HTML	-	HyperText Markup Language
HTTP	-	HyperText Transfer Protocol
JSON	-	JavaScript Object Notation
LLM	-	Large Language Model
LSTM		Long Short-Term Memory
MCA	-	Master of Computer Applications
NSFW	-	Not Safe For Work
ORM	-	Object-Relational Mapping
PDF	-	Portable Document Format
PO	-	Program Outcome
POST	-	POST Request Method
PSO	-	Program Specific Outcome
RNN	-	Recurrent Neural Network
SDG	-	Sustainable Development Goal
SQL	-	Structured Query Language
UI	-	User Interface
UML	-	Unified Modeling Language

CHAPTER 1
INTRODUCTION

1.1 Background

Social media has become an important part of everyday communication. People use social networking platforms to share photos, videos, messages, opinions, and other forms of personal content. These platforms have changed the way users communicate by making it possible to interact with friends, communities, and other users through a single online environment. As the number of social media users continues to grow, there is an increasing need for platforms that provide useful communication features while also giving users better control over their content and personal information.

Many existing social networking applications provide a large collection of features, including user profiles, multimedia posts, stories, short videos, messaging, comments, likes, notifications, and search facilities. However, these features are generally provided through large external platforms where users have limited control over the underlying application environment and data management. Building a smaller, locally deployable social networking system provides an opportunity to understand how these features can be integrated into a single web application.

NILA SOCIAL MEDIA is developed as a web-based social networking application that combines common social interaction features with selected AI-assisted functions. The system allows users to create accounts, manage profiles, share multimedia content, follow other users, interact through likes and comments, communicate through messages, and receive notifications. Privacy-related features such as account controls and follow management are also included to provide users with greater control over their social interactions.

The project also explores the use of artificial intelligence within a social networking environment. AI-assisted functionality can be used to support selected activities such as content assistance, recommendation-related processing, and image or content analysis. Instead of treating AI as a separate component, the project considers how such functionality can be integrated with the normal workflow of a social media application.

NILA SOCIAL MEDIA therefore provides a practical implementation of a social networking platform while also demonstrating the integration of web application

development with AI-assisted features. The modular structure of the system makes it easier to maintain the different functions and provides a foundation for future improvements and additional features.

1.2 Motivation

The rapid growth of social networking has made online platforms an important part of communication and content sharing. Users now expect social media applications to provide multiple services such as posting content, communicating with others, receiving notifications, and managing their privacy within a single platform. This encouraged the development of a project that combines these commonly used features into a structured and manageable web application.

Another motivation for the project is the growing use of artificial intelligence in web-based applications. AI can assist users in performing certain tasks and can also be used to analyse content and user interactions. Integrating selected AI-assisted functions into a social networking application provides an opportunity to explore how intelligent technologies can be applied to improve the overall functionality of a web system.

The project was also motivated by the need to understand the development of a complete full-stack application rather than implementing individual features separately. Designing user authentication, database operations, multimedia handling, social interactions, messaging, notifications, and privacy controls within one system provides practical experience in software development, database management, web technologies, and AI integration.

NILA SOCIAL MEDIA was therefore developed as an academic project to demonstrate the practical application of these technologies in a single platform. The project provides a controlled environment for developing and testing social networking features while creating a foundation for future improvements in intelligent content assistance, recommendations, security, and scalability.

1.3 Scope & Objectives

Scope

The scope of NILA SOCIAL MEDIA covers the development of a web-based social networking platform that brings commonly used communication and content-sharing functions together in one application. The system is designed to support users in creating

accounts, maintaining profiles, sharing multimedia content, and interacting with other users.

The application includes features such as posts, stories, reels, likes, comments, follows, messaging, notifications, and search. Privacy-related functions are also considered so that users can manage their accounts and control certain interactions with other users. Administrative functions can be used to manage and monitor the application according to the requirements of the system.

The project also covers the integration of selected AI-assisted functionality into the social media environment. These functions can support areas such as content assistance, recommendation-related processing, and image or content analysis. The AI components are intended to complement the main application rather than replace its basic social networking functions.

The current project is primarily intended for academic development, local testing, and demonstration. Its modular structure provides scope for future improvements such as enhanced AI models, better recommendation mechanisms, improved scalability, additional multimedia features, and deployment on larger environments.

The system also provides a practical environment for studying the interaction between the frontend, backend, database, and supporting services of a social networking application. The different modules can be developed and maintained separately, making it easier to identify issues, update individual components, and introduce additional functionality without affecting the entire system.

Another part of the scope is to provide a foundation for experimenting with intelligent features within a familiar social media environment. The project can be used to study how user interactions and digital content can be handled by web-based and AI-assisted components. This makes the system useful not only as a social networking application but also as an academic platform for understanding the integration of modern software technologies.

Objectives

The main objectives of NILA SOCIAL MEDIA are:

1. To develop a web-based social networking platform that provides essential communication and content-sharing features in a single application.
2. To provide secure user authentication and profile management so that users can create and maintain their personal accounts.
3. To support multimedia content sharing through posts, stories, and reels.
4. To enable social interaction through likes, comments, follows, and other user engagement features.
5. To provide communication facilities such as direct messaging and notifications.
6. To include privacy and account-control features that allow users to manage their social interactions and content visibility.
7. To integrate selected AI-assisted functions for supporting content-related and recommendation-related activities.
8. To provide search and exploration facilities that help users discover relevant content and other users.
9. To design the application using a modular structure so that individual components can be maintained and improved independently.
10. To provide a foundation for future development by allowing additional AI capabilities, scalability improvements, and new social networking features to be incorporated into the system.

1.4 Problem Statement

The widespread use of social networking platforms has created a demand for applications that can provide communication, content sharing, and social interaction through a single system. Modern users expect features such as profile management, multimedia sharing, messaging, notifications, search, and privacy controls to be available within the same

platform. Developing and integrating these different functions into one application requires proper coordination between the user interface, application logic, database, authentication system, and media-handling components.

Another challenge is the integration of intelligent functionality into social networking applications. Artificial intelligence can be used to assist with selected activities such as content analysis, recommendations, and other user-support functions. However, incorporating such features into a conventional web application requires suitable integration with the existing application structure while maintaining usability and system reliability.

There is also a need to provide users with greater control over their accounts and social interactions. Features such as private accounts, follow management, content visibility, and user-related controls need to be considered as part of the overall system design rather than being treated as separate functions.

Therefore, the problem addressed by this project is the development of a structured and modular social networking web application that combines essential social media functions, communication facilities, privacy controls, and selected AI-assisted features within a single system. NILA SOCIAL MEDIA aims to address this requirement through an integrated web-based platform that can be developed, tested, and demonstrated in a controlled environment.

CHAPTER 2

REQUIREMENT AND ANALYSIS

2.1 Existing System and Problem Statement

Existing System

Existing social networking platforms provide users with a wide range of facilities for communication, content sharing, and online interaction. Common functions include user registration, profile management, multimedia posts, likes, comments, following, stories, short-form videos, private messaging, notifications, and search. These platforms demonstrate how different social features can be integrated into a single application and accessed through web or mobile interfaces.

Although existing platforms provide many useful features, they are generally designed as large-scale commercial systems with complex infrastructure and services operating behind the user interface. For an academic project, working with such platforms does not provide direct control over the application architecture, database structure, feature implementation, or processing workflow. This makes it difficult to study the complete development process of a social networking system in a controlled environment.

Another aspect is the integration of artificial intelligence into social applications. AI-based techniques can assist with content-related tasks, recommendations, and image or text analysis. However, these functions need to be properly connected with the existing user, content, and database modules. A suitable application structure is therefore required to demonstrate how AI-assisted components can work together with conventional web application features.

Privacy and user control are also important considerations in a social networking environment. Users may require options for managing their profiles, controlling interactions, and restricting access to certain content or activities. These requirements need to be considered during system design so that privacy-related functions are integrated with the main application rather than added as independent features.

Problem Statement

The main problem addressed by NILA SOCIAL MEDIA is the development of a structured web-based social networking application that combines essential social interaction features with selected AI-assisted functionality in a manageable environment. The system needs to coordinate user authentication, profile management, multimedia content, social

interactions, messaging, notifications, search, privacy controls, and administrative functions.

The application must also maintain proper communication between its frontend, backend, database, and AI-assisted components. User data and application content need to be handled through an organized database structure, while different modules should work together without unnecessary dependency between them.

Therefore, the proposed system focuses on developing **NILA SOCIAL MEDIA** as a modular social networking platform that can be developed, tested, and demonstrated in an academic environment. The system aims to provide the required social features while providing a foundation for integrating AI-assisted functionality and future improvements.

2.2 Solution Overview

NILA SOCIAL MEDIA is proposed as a web-based social networking application that brings user management, content sharing, communication, and social interaction into a single platform. The system is designed with separate modules so that each major function can be developed and managed in an organized manner. This approach helps the application maintain a clear structure while allowing individual components to be improved when required.

The system begins with user registration and authentication. After creating an account, users can maintain their profiles and interact with other users through the available social features. Users can create and share posts, stories, and reels, while other users can respond through likes and comments. The follow mechanism allows users to connect with other accounts, and privacy-related controls can be used to manage selected interactions and account visibility.

Communication is supported through direct messaging and notifications. The search and exploration facilities help users find accounts and available content within the application. Administrative functions provide a way to manage and monitor relevant system activities according to the requirements of the application.

An important part of the proposed solution is the integration of selected AI-assisted functions. These functions are connected with the main application to support activities such as content assistance, recommendation-related processing, and image or content analysis. The AI components are treated as supporting services within the system and are designed to work alongside the normal social networking modules.

The application follows a structured flow between the user interface, application logic, database, media-handling components, and AI-assisted services. User requests are

processed by the application, relevant information is stored or retrieved from the database, and the appropriate result is presented through the interface. This organization helps maintain consistency between different modules and simplifies future maintenance.

Overall, the proposed solution provides a controlled environment for implementing and studying the major components of a social networking application. The modular design also allows the system to be extended in the future with improved AI functionality, additional social features, enhanced security, and larger-scale deployment.

2.3 Feasibility Study

A feasibility study is carried out to determine whether the proposed **NILA SOCIAL MEDIA** system can be developed and used successfully within the available resources and project requirements. It helps to identify possible technical, financial, operational, and time-related challenges before completing the implementation. The study indicates whether the proposed system is practical for academic development and demonstration.

2.3.1 Technical Feasibility

NILA SOCIAL MEDIA is technically feasible because the system can be developed using commonly available web development technologies and programming tools. The application uses a Python-based backend framework, a database for storing user and application information, and standard web technologies for the user interface. These technologies provide the necessary support for implementing authentication, profiles, posts, social interactions, messaging, notifications, and other application modules.

The project can also incorporate selected AI-assisted functionality through suitable Python libraries and supporting components. Since the application is organized into separate modules, AI-related functionality can be connected with the required parts of the system without changing the complete application structure. Development and testing can be carried out on a standard computer suitable for running the required software environment.

2.3.2 Economic Feasibility

The project is economically feasible because its development can be carried out using commonly available development tools and software libraries. Many of the technologies required for the application are freely available and can be used for academic development without requiring expensive commercial software.

The project is primarily intended for academic purposes, local testing, and demonstration. Therefore, the initial development does not require a large-scale production infrastructure. The modular design also allows additional resources to be introduced later if the system is extended for wider deployment.

2.3.3 Operational Feasibility

The proposed system is operationally feasible because its main functions are based on familiar social networking activities. Users can register, log in, manage profiles, share content, interact with other users, send messages, and receive notifications through a web interface.

The modular organization of the system also makes it easier to manage different operations separately. Administrative functions can support the monitoring and management of relevant activities. The interface is designed to provide a straightforward interaction flow so that users can access the available features without requiring specialized technical knowledge.

CHAPTER 3

LITERATURE REVIEW

3.1 : Automated Image Captioning (Puscasiu et al., 2020)

This paper presents an image captioning approach built on an encoder-decoder design, in which a convolutional neural network (CNN) extracts visual features from the image and a recurrent neural network (RNN) generates the caption one word at a time. The authors use transfer learning, reusing a CNN pretrained on a large image dataset, so that good results can be obtained without training the whole network from scratch. The model is evaluated with BLEU and METEOR and is reported to produce accurate captions.

Relevance to this project: This paper represents the classical CNN-RNN baseline from which modern captioning models evolved. It gives Nila Social's AI layer a starting point for explaining why a pretrained model is reused for caption generation instead of being trained from scratch on a small classroom dataset.

3.2: Image Captioning: A Survey (Sharma et al., 2020)

This survey reviews deep-learning-based image captioning methods and compares them on common datasets such as MS COCO and Flickr30k and on standard evaluation metrics. It traces how encoder-decoder models and attention mechanisms improved caption quality over earlier template-based and retrieval-based approaches.

Relevance to this project: The survey gives the project a broad map of the captioning field and supports the choice of a deep-learning approach for the Local AI Studio. It also explains the datasets and metrics used in Chapter 3 when comparing the other captioning papers.

3.3: BLIP — Bootstrapping Language-Image Pre-training (Lietal., 2022)

BLIP is a vision-language pre-training framework that handles both understanding and generation tasks in a single model. It uses a bootstrapping strategy in which a captioner generates synthetic captions for web images and a filter removes noisy ones, which cleans up the large, noisy image-text pairs collected from the web. It is evaluated with metrics such as CIDEr and BLEU and shows better caption quality, along with strong results on image-text retrieval.

Relevance to this project: Nila Social's Local AI Studio uses the pretrained Salesforce/blip-image-captioning-base checkpoint described in this paper to generate captions for uploaded

posts. It is the most important piece of related work for the project's AI layer, because the model is small enough to run locally without paid external APIs.

3.4: Deep Learning Approaches on Image Captioning: A Review (Ghandi et al., 2022)

This review organises modern deep-learning captioning models into categories such as CNN-RNN models, attention-based models and transformer-based models. It discusses the strengths and limitations of each category and compares them using metrics such as BLEU and CIDEr.

Relevance to this project: The review helps justify why a transformer-based model was selected for Nila Social over older CNN-RNN designs. It also provides a structured way to describe the limitations of the chosen approach in the Limitations discussion of this report.

3.5: Review of Transformer-Based Image Captioning Models (Ondeng et al., 2023)

This paper reviews captioning models built on the Transformer architecture, which replaces recurrence with self-attention. Because self-attention allows parallel processing and captures long-range dependencies better than RNNs, transformer-based captioners achieve stronger performance on BLEU and CIDEr than earlier recurrent models.

Relevance to this project: BLIP, the model used in Nila Social, is built on the transformer architecture, so this review explains why the captioning model in this project performs well. It supports the technical background section on how the model attends jointly over image regions and generated text.

3.6: Captioning by Diffusion Image Models: A Survey (Daneshfar et al., 2024)

This survey examines the use of diffusion models for image captioning. Diffusion models, which became well known for image generation, produce output by gradually removing noise, and applied to captioning they allow a caption to be refined as a whole instead of being generated strictly word by word. The survey compares these approaches using BLEU and CIDEr and presents them as one of the latest directions in the field.

Relevance to this project: Nila Social does not use diffusion-based captioning, since such models are heavier to run in an offline, classroom-scale deployment. The paper is cited to show awareness of newer approaches and is listed as possible future work for the AI Studio.

3.7: Attention-Based Transformer Models for Multilingual Image Captioning (Albadarneh et al., 2025)

This paper applies attention-based transformer models to generate image captions in multiple languages instead of English alone. The attention mechanism helps the model capture the relevant image regions and the language context, and the work is evaluated with BLEU and ROUGE, showing better contextual captions.

Relevance to this project: The paper supports the bilingual caption feature of the Local AI Studio, which aims to serve users beyond English. It provides academic grounding for treating multilingual captioning as a meaningful feature for a regional social media application.

3.8: Next-Generation Image Captioning: A Systematic Review (Al-Malla et al., 2026)

This systematic review analyses recent captioning research built on vision-language models. It compares work using BLEU, CIDEr and SPICE, where SPICE evaluates the semantic correctness of objects, attributes and relations in a caption, and it outlines future directions for the field such as larger vision-language models, better evaluation methods and wider language support.

Relevance to this project: The review places Nila Social's BLIP-based captioning within the current direction of the field. It is also used in the Future Scope discussion, where upgrading to a larger vision-language model and a richer evaluation is identified as a natural next step.

3.11 Summary of Literature Review

	Theme	Key Idea Adopted in Nila Social
1	Vision-language pretraining (BLIP)	Local captioning model for the Local AI Studio

	Theme	Key Idea Adopted in Nila Social
2	CNN-RNN encoder-decoder with transfer learning	Basis for resuing a pretrained model instead of training a captioner from scratch
3	Survery of deep learning captioning approaches	Overview of datasets and metrices justification for a deep-learning approach
4	Review of deep learning captioning approaches	Comparison of CNN-RNN, attention and transformer models, basis for the limitations discussion
5	Transformer based captioning	Backbone understanding for why the captioning model performs well
6	Diffusion -based captioning	Identified as future work for the AI Studio
7	Mutltilingual attention based captioning	Academic support for the bilingual caption feature
8	Systematic review of vision-language models	Positions the project in the current filed; guides future scope

Table 3.1: Literature Review Summary

3.12 Research Gap

Although existing studies have significantly improved image captioning using CNN-RNN, attention, Transformer, BLIP, multilingual, and diffusion-based approaches, several gaps remain:

- Most existing models focus mainly on caption quality and benchmark performance, rather than practical social-media applications.
- Many advanced vision-language models require high computational resources or cloud-based services, making them less suitable for local/offline use.
- Multilingual caption generation, especially for regional languages, is still less explored compared with English captioning.
- Existing research mainly uses standard datasets like MS COCO and Flickr30k, which may not reflect real-world social-media images.
- There is limited focus on integrating automatic caption generation directly into a lightweight social-media platform.
- Recent approaches such as diffusion-based and larger vision-language models are powerful but can be too resource-intensive for classroom-scale/local deployment.

CHAPTER 4
SYSTEM DIAGRAMS

4.1 Data Flow Diagrams (DFD)

This chapter presents the system-modelling diagrams for Nila Social: data flow diagrams (context and level-1), an entity-relationship diagram, a use case diagram, a class diagram, a state diagram, an activity diagram and a sequence diagram. Every diagram was generated directly from the implemented Flask blueprints, SQLAlchemy models and service modules described in Chapters 5 and 6, rather than drawn independently of the code, so that the diagrams and the implementation remain consistent.

4.1.1 DFD Level 0 (Context Diagram)

The Level-0 (context) diagram treats Nila Social as a single process exchanging data with two external entities — the User and the Admin — and with its persistent store (SQLite database and media file storage).

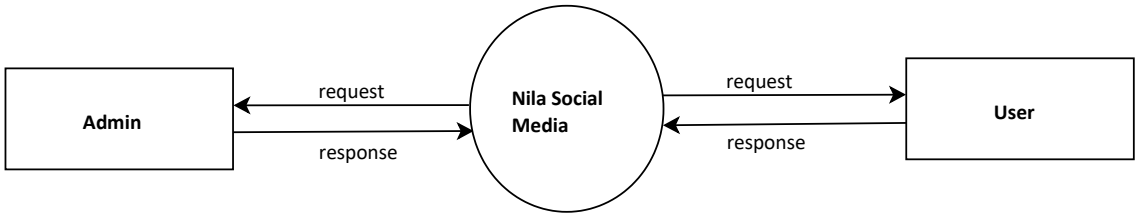


Figure 4.1: DFD Level 0 (Context Diagram)

4.1.2 DFD Level 1

The Level-1 diagram decomposes the system into seven major processes — Authentication, Content Management, Social Graph, Engagement, Messaging, Local AI Services, and Moderation & Reporting — each reading from and writing to its own data store. Content Management and Engagement both invoke the Local AI Services process (for captioning/tagging and for comment screening respectively), which is the key integration point between the "conventional" CRUD part of the system and its AI layer.

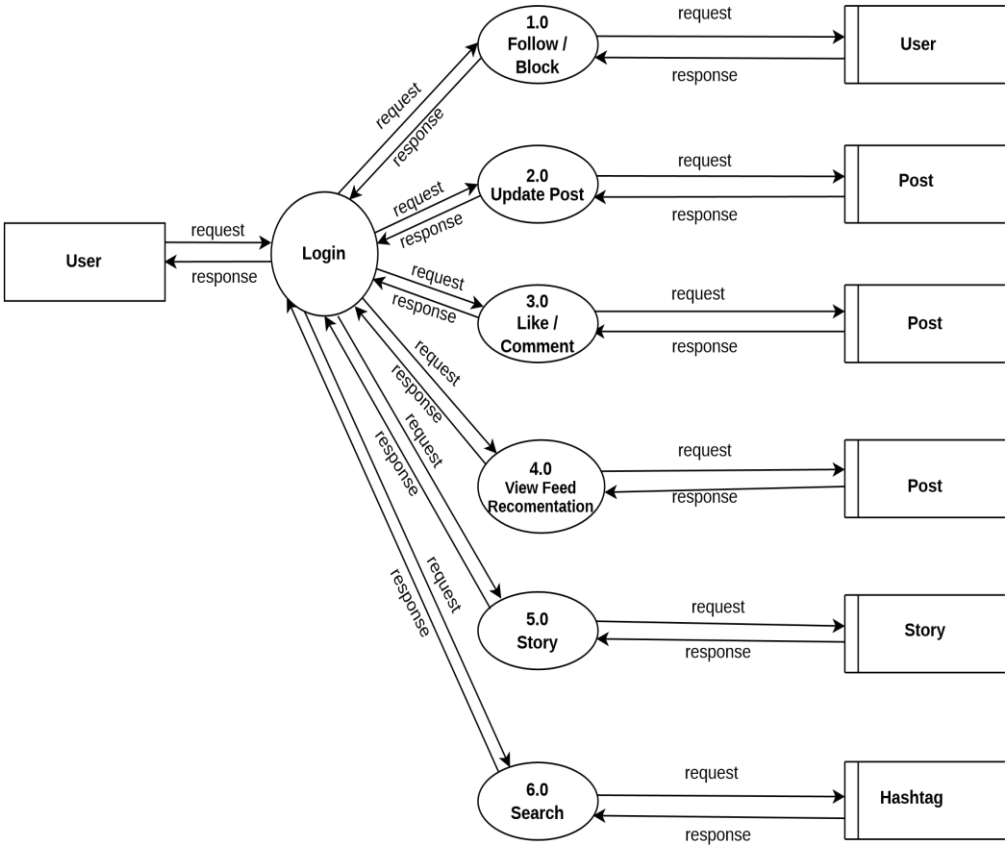


Figure 4.2: DFD Level 1

4.3 Use Case Diagram

Four actors interact with the system: a Guest (unauthenticated visitor), a Registered User, a Private-Account User (a specialisation of Registered User that must additionally approve follow requests), and an Admin. The Create Post/Story/Reel use case includes the Local AI Studio use case (every media upload can optionally invoke AI captioning), and the Report Content use case extends into the Admin's Moderate Reports use case.

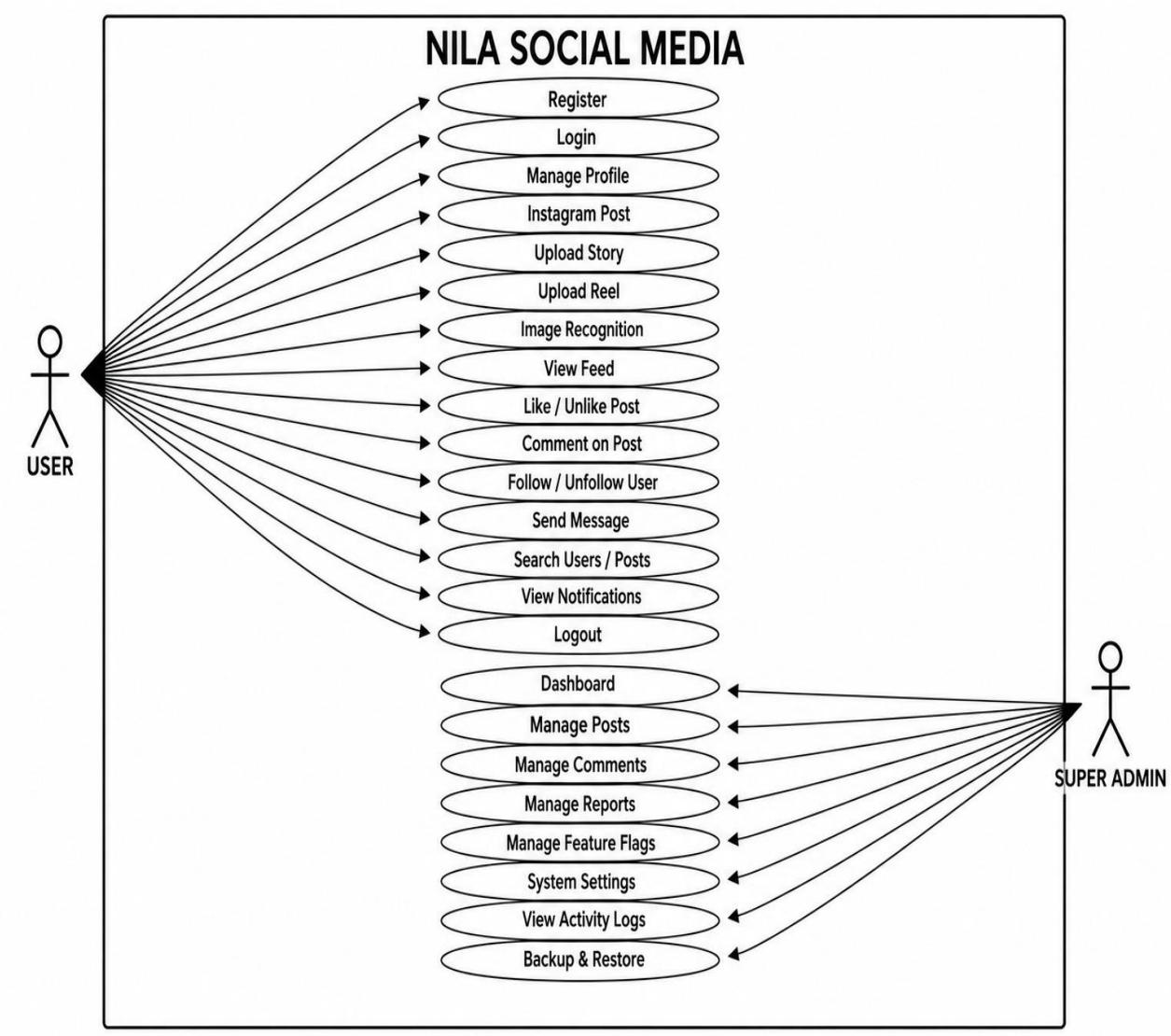


Figure 4.4: Use Case Diagram

4.4 Class Diagram

The class diagram summarises the principal SQLAlchemy model classes and their associations, alongside the AIEngine service class that Post and Comment depend on for AI-assisted captioning and moderation. Attribute visibility follows UML convention (+ public, − private); methods shown are illustrative of the behaviour implemented on each model (e.g., User.set_password(), Post.like_count(), Follow.accept()).

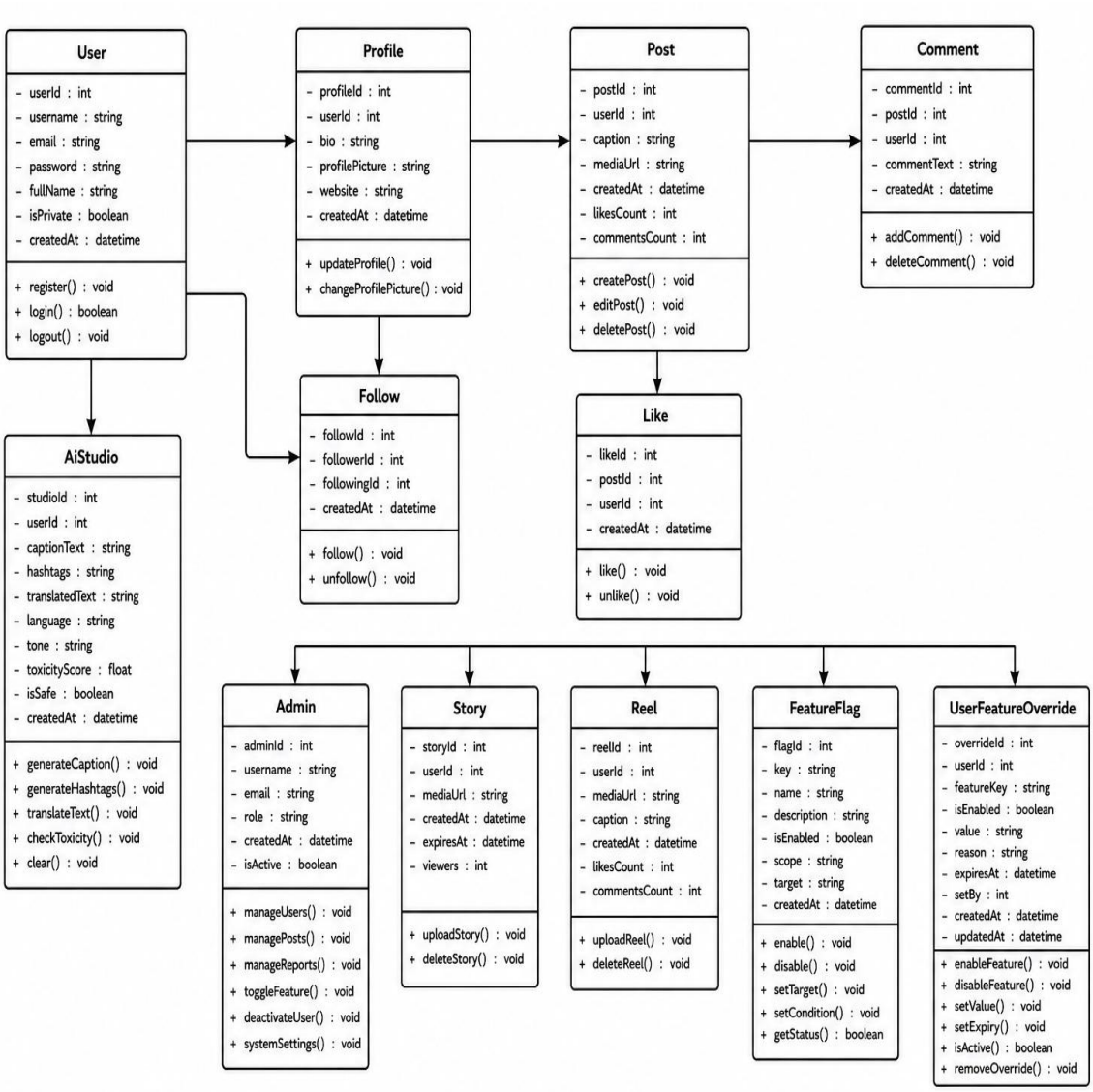


Figure 4.5: Class Diagram

4.5 State Diagram

The state diagram models the lifecycle of a Post from creation to removal. A newly created post first passes through AI Processing (caption/tag generation and moderation scanning). If the moderation check fails (NSFW or toxic caption), the post is Flagged for admin review rather than immediately published; otherwise it becomes Published and visible in followers' feeds. From Published, the owner may Archive (hide without deleting) or Delete the post; a Flagged post is resolved by an admin either back to Published or to Removed.

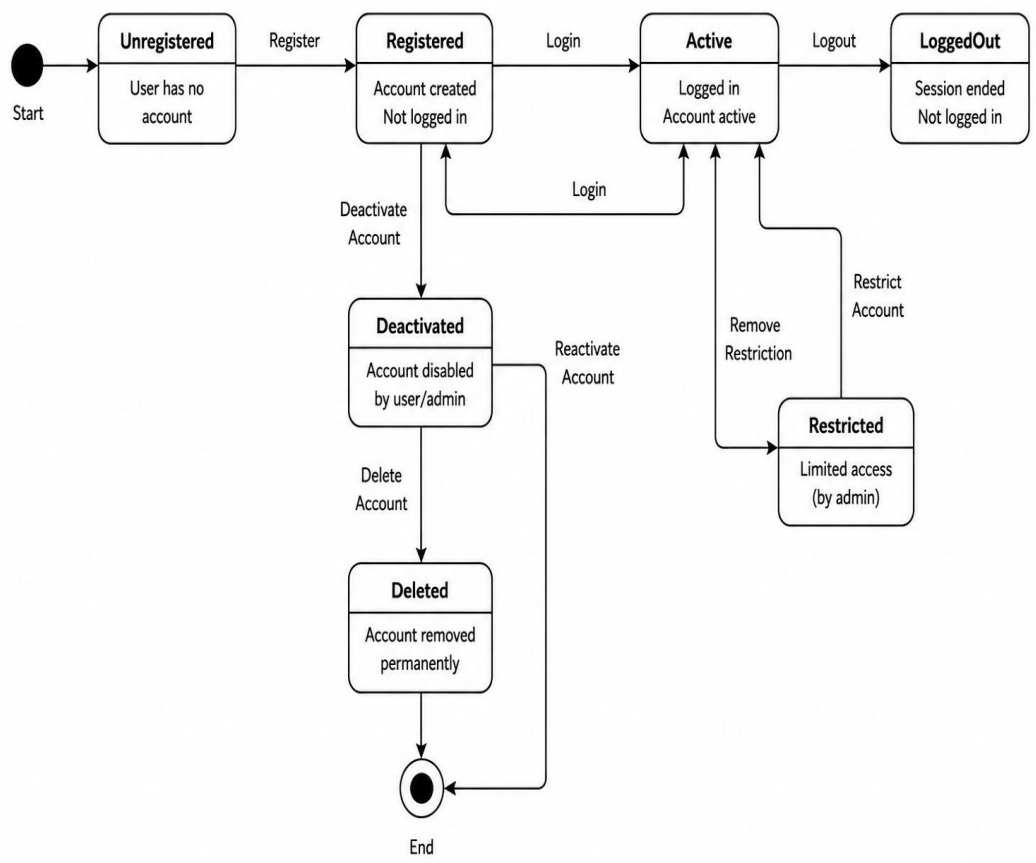


Figure 4.6: State Diagram — Post Lifecycle

4.6 Activity Diagram

The activity diagram traces the create-post workflow end-to-end: the user selects media, the file is uploaded to storage, the user optionally invokes the Local AI Studio to generate a caption/hashtags, the user finalises caption/location/audience settings, the moderation service scans the combined text and image, and the post is either saved and published to followers' feeds or flagged for admin review.

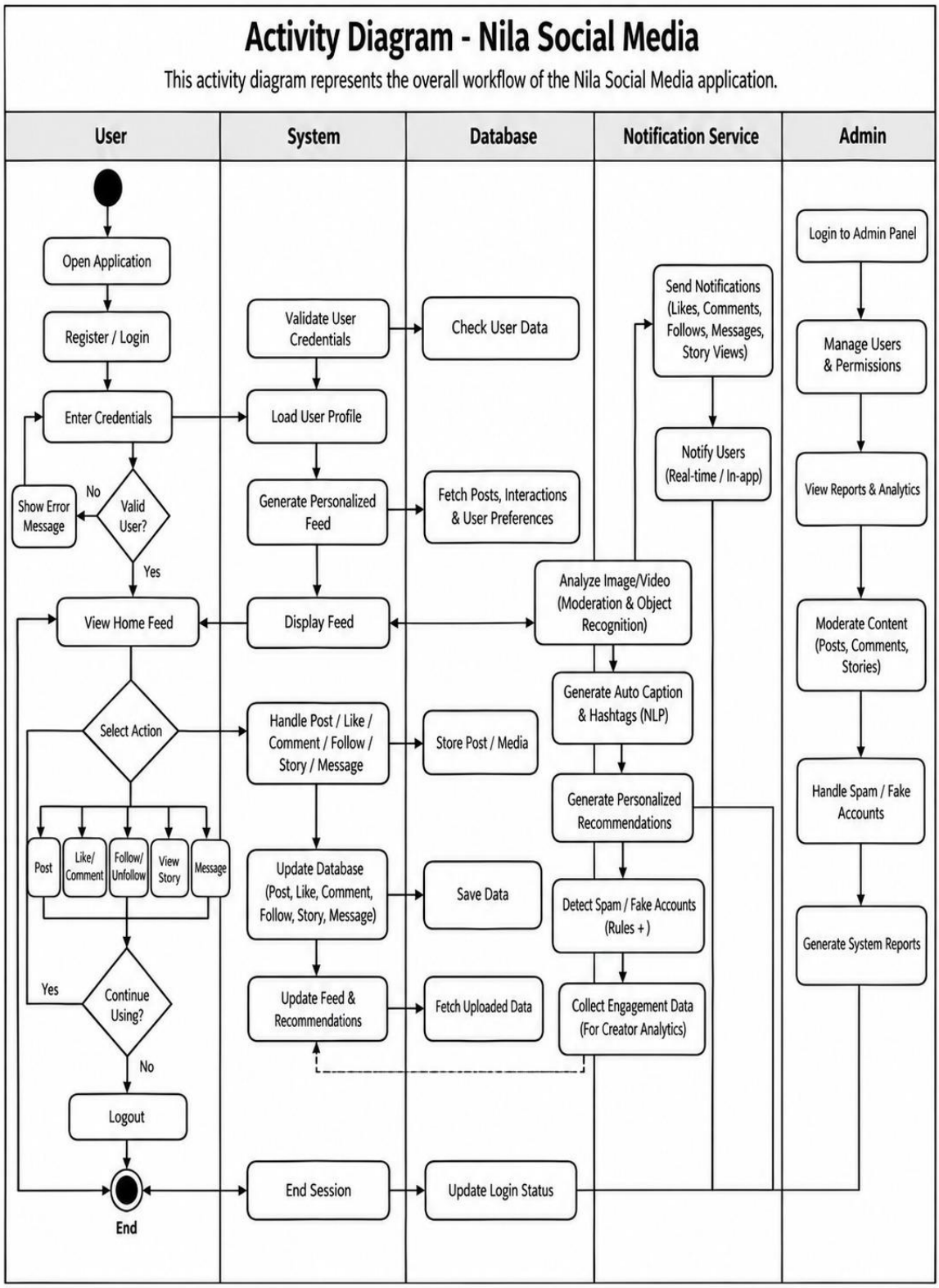


Figure 4.7: Activity Diagram — Create Post with AI Assist

4.7 Sequence Diagram

The sequence diagram below traces a login request across the five components involved: the login form, the auth.login Flask route, the SQLAlchemy User model, the SQLite database, and the Flask-Login session manager. This same request/response pattern (route → ORM model → database → route → template) recurs throughout the system and is representative of how every blueprint in Table 5.1 is structured internally.

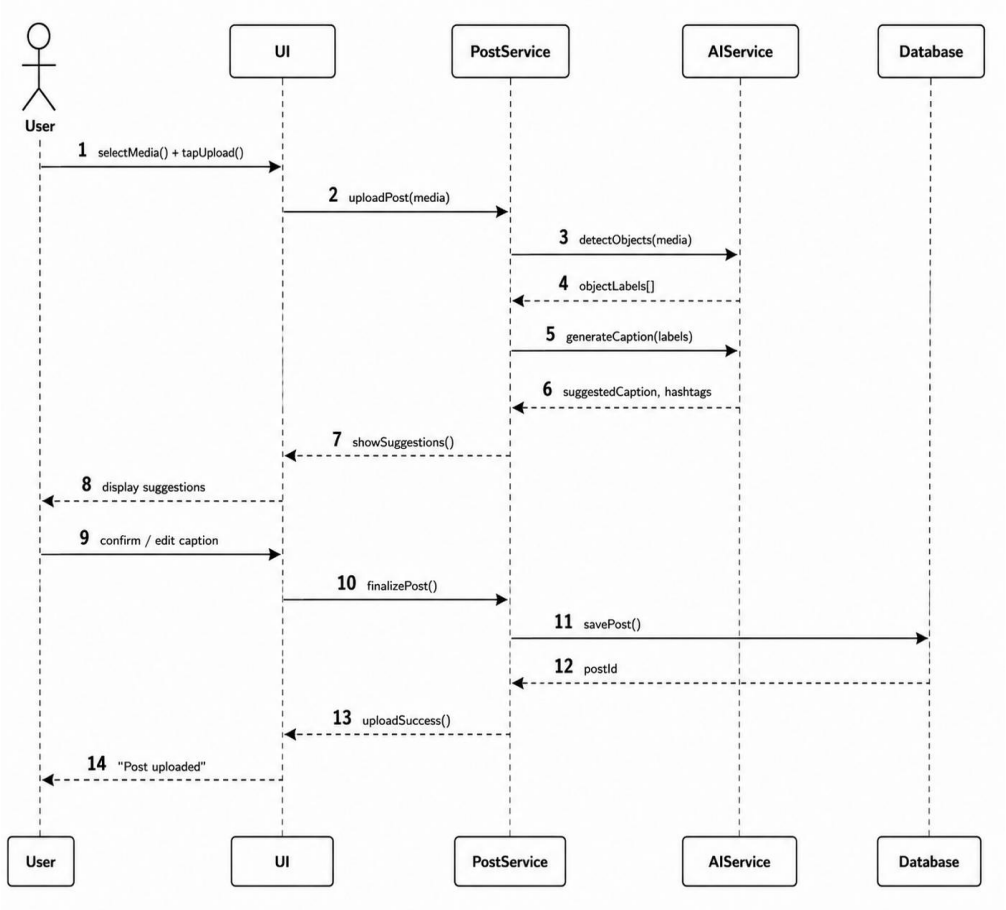


Figure 4.8: Sequence Diagram — Login Flow

CHAPTER 5
METHODOLOGY

5.1 Module Descriptions

NILA SOCIAL MEDIA is divided into a number of functional modules, with each module handling a specific part of the application. The modules work together to provide users with a complete social networking experience. User authentication and profile management form the basic foundation of the system, while content sharing, social interaction, communication, privacy, and AI-assisted functions provide the main application features. The modular structure also makes the system easier to understand, test, maintain, and extend.

5.1.1 Landing, Registration and Login

An unauthenticated visitor lands on a marketing-style landing page; existing users log in with an email/password pair validated against a Werkzeug-hashed password_hash column, and new users register through a WTForms-validated form (unique username/email, password confirmation) before being routed into a profile-setup step.

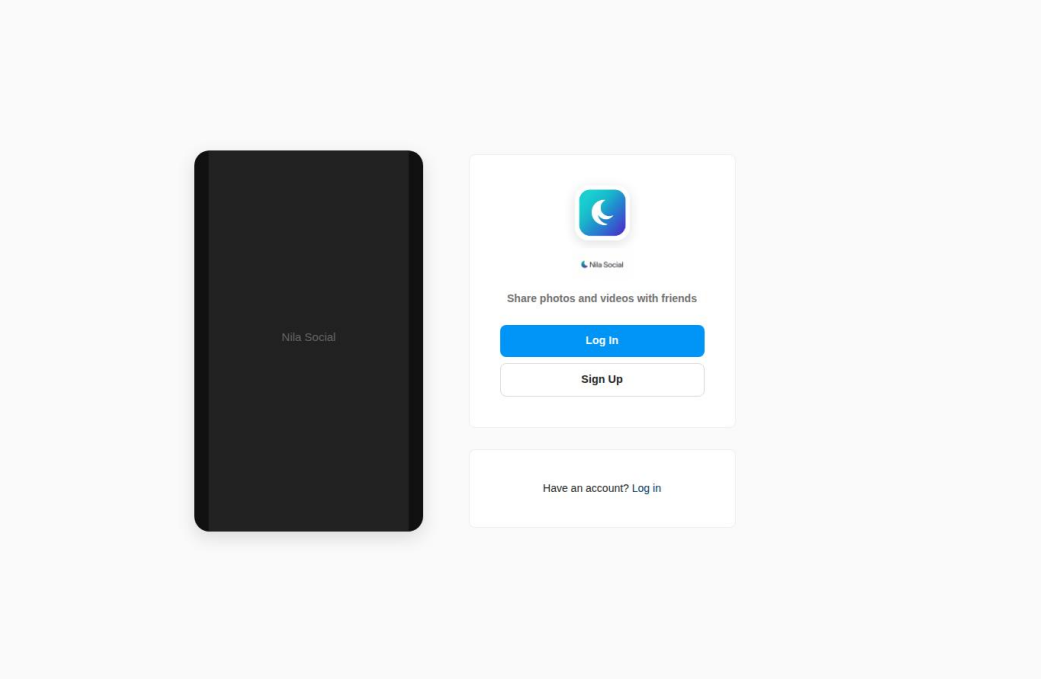


Figure 5.1: Landing Page

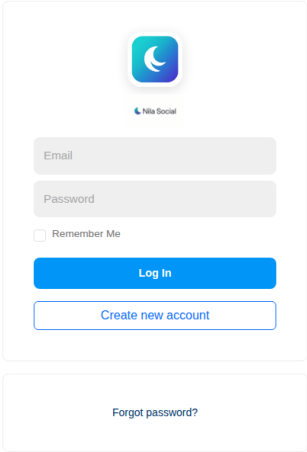


Figure 5.2: Login Page

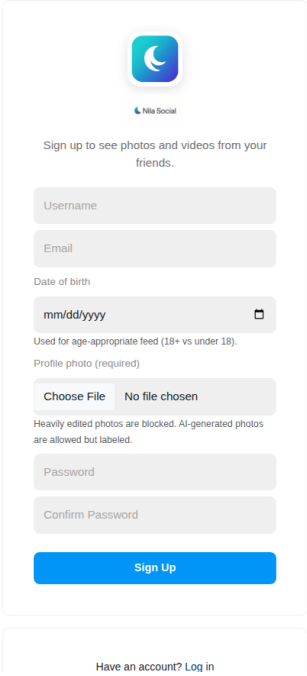


Figure 5.3: Registration Page

5.1.2 Home Feed

After login, GET /feed renders a stories tray (24-hour, non-expired stories from accounts the user follows), a horizontally scrollable reels tray, and the ranked post feed produced by get_recommended_posts() in app/services/ai_feed.py (see Section 5.2 for the ranking formula). A "Suggested for you" panel lists accounts the user does not yet follow.

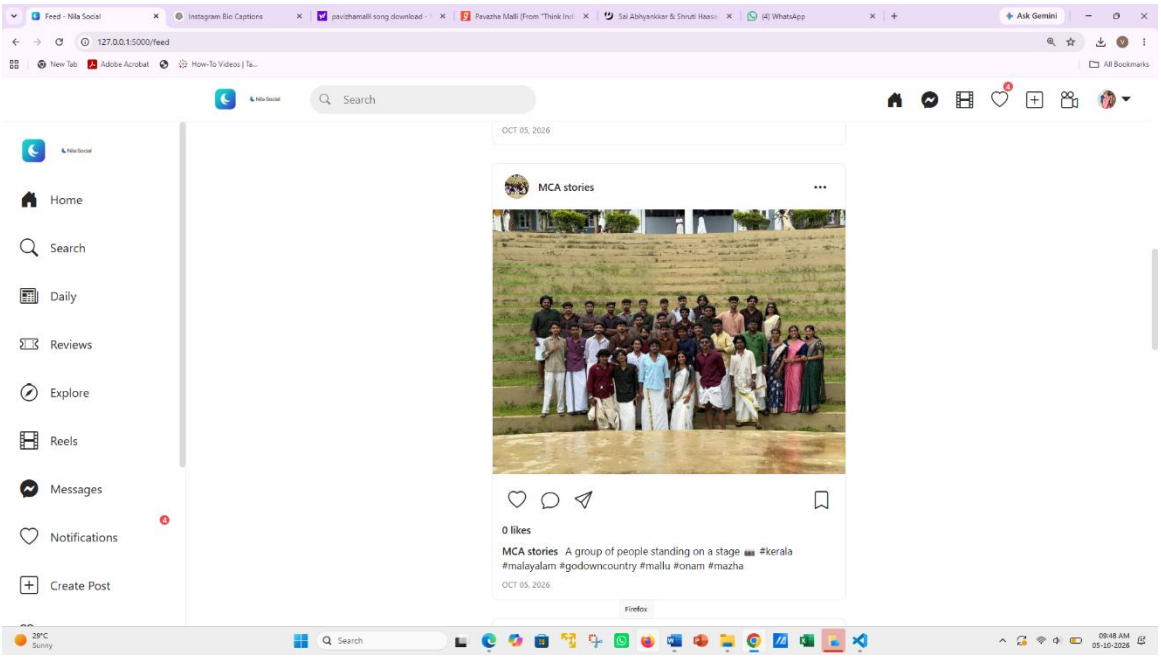


Figure 5.4: Home Feed with Stories and Reels Tray

5.1.3 Create Post and the Local AI Studio

The Create Post page (GET/POST /post/create) is the most feature-dense screen in the system: alongside the standard file picker, caption text area and location field, it embeds the Local AI Studio panel, which exposes 41 distinct AI-assisted actions grouped under headings such as "Caption from Image", "Hashtags from Image", "Full Photo Analysis", "Suggest Location (CLIP)", "Image Quality Check", "Engagement Predictor", "Alt Text + Styles" and "Find Similar Posts". Each button issues an asynchronous POST request to the corresponding endpoint in app/routes/api.py (e.g. /api/ai/caption, /api/ai/hashtags, /api/ai/quality), which is handled by the matching ai_*.py service module described in Section 5.2. The panel header text ("Local AI Studio — fallback · 41 features") is generated dynamically by /api/ai/status and reports whether the transformer-backed BLIP model or the rule-based fallback is currently active, which is a deliberate transparency feature so a user (or evaluator) always knows which mode produced a given suggestion.

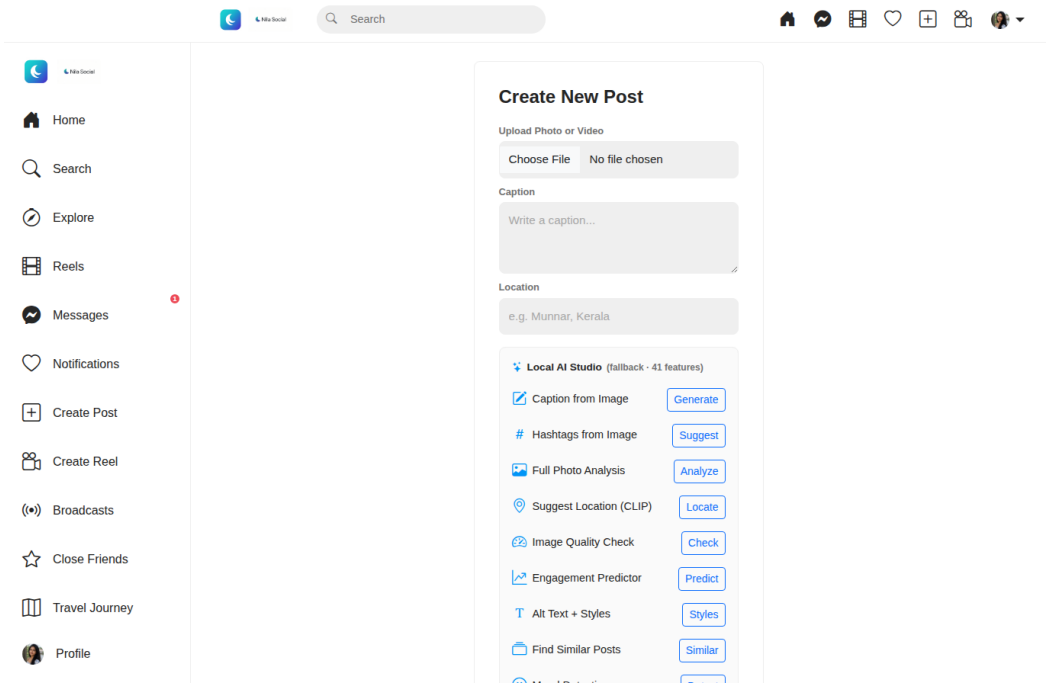


Figure 5.5: Create Post with Local AI Studio (41 Features)

5.1.4 Profile

The profile page (GET /profile/<username>) renders the post/reel/saved/tagged tabs, follower/following counts, and — for the profile owner — an Edit Profile action; for a private account viewed by a non-follower, the profile.private template is rendered instead, and no post content is served, enforcing privacy at the route level rather than only visually.

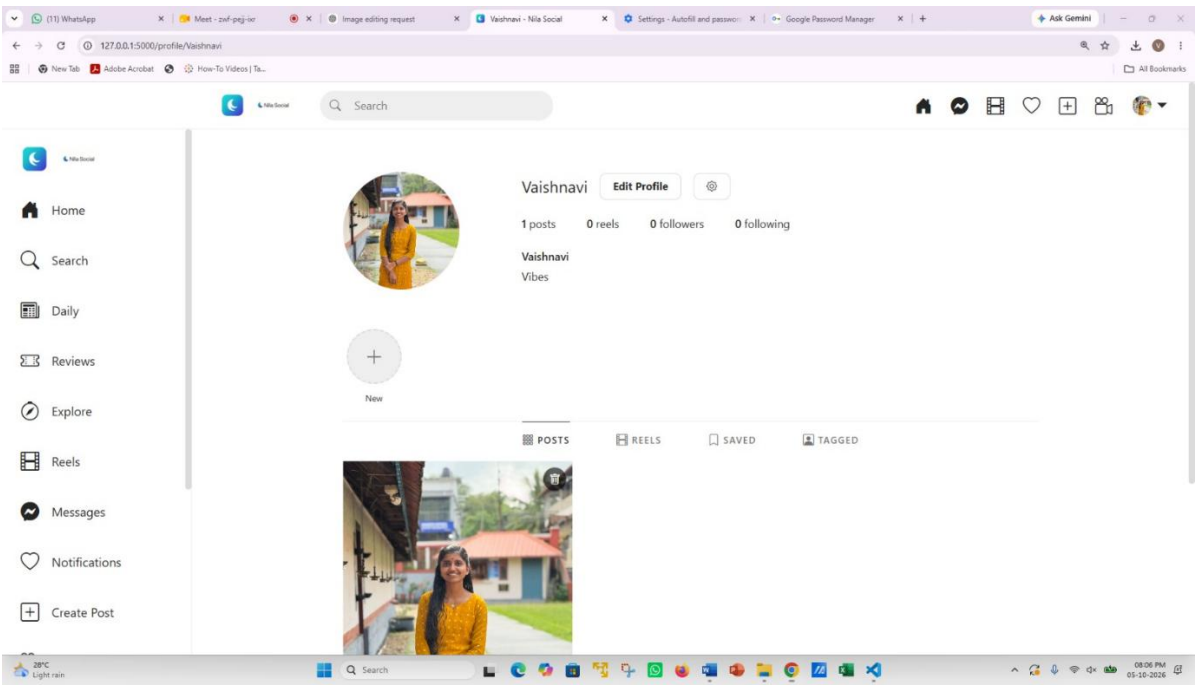


Figure 5.6: User Profile Page

5.2 Algorithms

NILA SOCIAL MEDIA uses a combination of application-level algorithms and processing workflows to manage user accounts, social interactions, multimedia content, privacy controls, and selected AI-assisted functions. The algorithms are implemented as part of the corresponding modules and work together with the database and backend application logic.

5.2.1 User Authentication Algorithm

The authentication algorithm verifies the identity of a user before allowing access to protected parts of the application.

Input: User login credentials

Process:

1. The user enters the required login credentials.
2. The system checks whether the account information is available.
3. The submitted credentials are validated against the stored account information.
4. If the credentials are valid, the user is authenticated.
5. A user session is created for the authenticated account.
6. If the credentials are invalid, access is denied and an appropriate message is displayed.

Output: Authenticated user session or login failure.

This algorithm ensures that user-specific functions are accessible only after successful authentication.

5.2.2 User Registration Algorithm

The registration algorithm creates a new user account after validating the information provided by the user.

Input: User registration details

Process:

1. The user enters the required registration information.
2. The system validates the submitted values.
3. The system checks whether the account information already exists.
4. If the information is acceptable, a new user record is created.
5. The account information is stored in the database.
6. The user can subsequently use the registered credentials to access the application.

Output: Successfully created user account or validation error.

5.2.3 Post Creation Algorithm

The post creation algorithm manages the process of publishing multimedia content.

Input: User account, media file, and caption/content information

Process:

1. The system verifies that the user is authenticated.
2. The submitted media is checked according to the supported file requirements.
3. The media is processed and stored using the application's media-handling mechanism.
4. The post information is associated with the authenticated user.
5. The post record is stored in the database.
6. The newly created post is made available through the appropriate application views.

Output: Successfully created post.

This process connects the authentication, media management, database, and post modules.

5.2.4 Like Processing Algorithm

The like algorithm manages a user's response to supported content.

Input: User account and content item

Process:

1. The system identifies the current user.
2. The selected content item is identified.
3. The system checks whether the user has already interacted with the content using the like function.
4. If the user has not liked the content, a like record is created.
5. If the user has already liked the content, the existing interaction can be removed according to the application's implementation.
6. The updated interaction information is stored in the database.
7. The relevant result is displayed to the user.

Output: Updated like status.

5.2.5 Comment Processing Algorithm

The comment algorithm allows users to add text-based responses to content.

Input: User account, selected content, and comment text

Process:

1. The system verifies the user's authenticated session.
2. The selected content is identified.
3. The submitted comment is validated.
4. The comment is associated with the user and selected content.
5. The comment record is stored in the database.
6. The updated comment information is displayed with the relevant content.

Output: Stored and displayed comment.

5.2.6 Follow Request Algorithm

The follow algorithm manages relationships between user accounts.

Input: Requesting user and target user

Process:

1. The system identifies the requesting user.
2. The target account is selected.
3. The system checks the privacy status of the target account.
4. If the account allows direct following, the relationship is created according to the application's rules.
5. If the account is private, the request is handled as a follow request and requires the appropriate approval.
6. The relationship or request is stored in the database.
7. The relevant user can receive a notification when applicable.

Output: Follow relationship or pending follow request.

5.2.7 Private Account Access Algorithm

This algorithm controls access to content and interactions associated with private accounts.

Input: Requesting user, target account, and requested resource

Process:

1. The system identifies the target account.
2. The privacy status of the account is checked.
3. If the account is public, the resource is made available according to normal access rules.
4. If the account is private, the system checks whether the requesting user has the required relationship or approval.
5. If permission exists, access is granted.

6. Otherwise, access to the restricted resource is denied.

Output: Access granted or access restricted.

5.2.8 Messaging Algorithm

The messaging algorithm manages the transfer and storage of messages between users.

Input: Sender, receiver, and message content

Process:

1. The sender is identified through the authenticated session.
2. The receiver is selected.
3. The message content is validated.
4. A message record is created with the relevant sender and receiver information.
5. The message is stored in the database.
6. The conversation is updated so that the participants can view the new message.
7. A notification can be generated where supported.

Output: Stored and displayed message.

5.2.9 Notification Processing Algorithm

The notification algorithm generates and displays relevant updates to users.

Input: Application event

Process:

1. An action occurs within the application.
2. The system identifies the users affected by the action.
3. The corresponding notification information is created.
4. The notification is associated with the appropriate user.
5. The information is stored in the database.
6. The notification is displayed through the user's notification area.

Output: Relevant notification displayed to the user.

5.2.10 Search Algorithm

The search algorithm helps users find accounts or content using a search query.

Input: Search keyword

Process:

1. The user enters a search term.
2. The system receives and validates the query.
3. Relevant records are searched using the supported searchable fields.
4. Matching results are retrieved from the database.

5. The results are presented to the user through the search interface.

Output: Matching users or content.

The search process provides a convenient way for users to locate information without manually browsing through the entire application.

5.2.11 AI-Assisted Content Processing

NILA SOCIAL MEDIA includes selected AI-assisted functionality to support content-related operations. The exact processing depends on the AI component implemented in the application.

Input: User-provided content or supported media

Process:

1. The relevant content is received by the AI-assisted component.
2. The input is prepared in the format required by the selected processing method.
3. The AI component analyses the input.
4. A result or supporting information is generated.
5. The result is returned to the application.
6. The application presents or uses the result in the corresponding feature.

Output: AI-assisted result.

The AI component is treated as a supporting part of the application. Normal social networking operations do not depend entirely on the AI component.

5.2.12 Recommendation Processing

The recommendation functionality is intended to use available user interaction information to support the discovery of relevant content or users.

Input: User interaction information and available content

Process:

1. The system identifies the current user.
2. Relevant interaction information is obtained from available application data.
3. The information is processed according to the recommendation logic implemented in the system.
4. Potentially relevant content or users are identified.
5. The selected results are presented to the user.

Output: Recommended content or users.

The recommendation process can be improved in future versions by incorporating more advanced machine-learning techniques and larger amounts of interaction data.

5.2.13 Media Upload Processing Algorithm

The media upload algorithm manages files submitted through supported content-sharing features.

Input: Uploaded media file

Process:

1. The system receives the uploaded file.
2. The file type and required upload conditions are checked.
3. The file is processed according to the application's media-handling rules.
4. The file is stored in the appropriate location.
5. A reference to the media is associated with the corresponding database record.
6. The media becomes available to the relevant application feature.

Output: Successfully processed media file.

5.2.14 Overall Application Processing Flow

The different algorithms work together as part of the complete application workflow. When a user performs an operation, the request first passes through the authentication and access-control mechanisms where required. The relevant module then processes the request, performs the necessary database operations, and returns the result to the user interface.

For example, when a user creates a post, the system verifies the user session, processes the uploaded media, creates the post record, stores the required information, and displays the resulting post. If another user interacts with that post, the corresponding interaction algorithm updates the database and can trigger a notification.

This approach allows the different algorithms to operate as individual components while still contributing to the overall functionality of NILA SOCIAL MEDIA. The modular implementation also makes it possible to improve individual algorithms in future versions without redesigning the entire application.

5.3 Architecture

The architecture of **NILA SOCIAL MEDIA** is designed to organize the different components of the application into separate but connected layers. The main purpose of this architecture is to provide a clear separation between the user interface, application

processing, data management, media handling, and AI-assisted functionality. This makes the system easier to develop, test, maintain, and extend.

The application follows a layered web application structure in which the user interacts with the system through the frontend interface. User requests are transferred to the backend application, where the required business logic is executed. The backend communicates with the database for storing and retrieving application data and interacts with supporting components whenever a particular service is required.

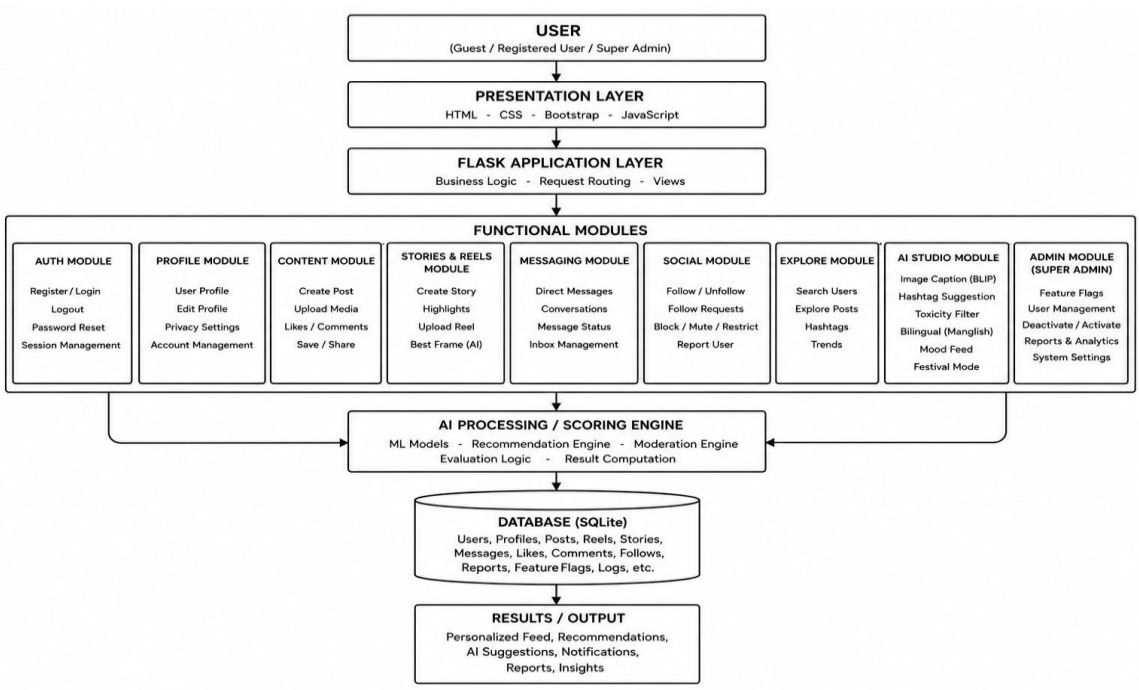


Figure 5.3: System architecture

5.3.1 Presentation Layer

The Presentation Layer forms the visible part of NILA SOCIAL MEDIA. It provides the interface through which users interact with the application.

This layer includes pages and interface components required for functions such as:

- User registration and login
- User profiles
- Home feed
- Post creation and viewing

- Stories
- Reels
- Likes and comments
- Follow and follow-request activities
- Messaging
- Notifications
- Search and explore
- Privacy-related settings
- Administrative functions

The presentation layer receives information from the backend and displays it in a user-friendly format. It also collects user input and sends requests to the application layer for processing.

5.3.2 Application / Business Logic Layer

The Application Layer acts as the main processing component of the system. It receives requests from the presentation layer and determines how each request should be handled.

This layer contains the logic associated with authentication, user management, content creation, social interactions, messaging, notifications, privacy controls, search, and other application functions.

For example, when a user creates a post, the application layer verifies the user's access, processes the submitted information, communicates with the media-handling component, and stores the required post information through the database layer.

Keeping the business logic separate from the user interface makes the application easier to maintain and reduces unnecessary dependency between individual components.

5.3.3 Authentication and Access Control

Authentication and access control form an important part of the backend architecture. Before accessing protected functions, the system identifies the user and verifies whether the required access conditions are satisfied.

The authentication component handles operations such as registration, login, logout, and session management. Access control is then used by other modules to determine whether a particular user is allowed to perform an operation or access a resource.

Privacy-related rules can also be applied at this stage. For example, access to content belonging to a private account can depend on the relationship or permission associated with the requesting user.

5.3.4 Data Management Layer

The Data Management Layer is responsible for communication between the application and the database. It stores and retrieves the information required by the different modules of NILA SOCIAL MEDIA.

The database can contain information related to:

- User accounts
- Profile information
- Posts
- Stories
- Reels
- Likes
- Comments
- Follow relationships
- Follow requests
- Messages
- Notifications
- Privacy settings
- Other application-related records

The backend communicates with the database through the appropriate data-access mechanism. This allows the application to maintain relationships between different records and retrieve the information required for each operation.

5.3.5 Media Management Layer

Multimedia content is an important part of NILA SOCIAL MEDIA. The Media Management component handles files associated with posts, stories, reels, profiles, and other supported features.

When a user uploads a file, the application checks and processes the upload according to the rules implemented in the system. The media is then stored using the application's media-storage mechanism, while the related information is associated with the corresponding database record.

This separation allows multimedia files and their database information to be managed in an organized manner.

5.3.6 AI Assistance Layer

The AI Assistance Layer provides selected intelligent functionality within the application. It is designed as a supporting component rather than as a replacement for the core social networking system.

Depending on the implemented AI features, this layer can be used for tasks such as:

- Content assistance
- Recommendation-related processing
- Image or content analysis
- Other selected AI-supported operations

When an AI-assisted operation is requested, the relevant input is passed from the application layer to the AI component. The AI component processes the input and returns the result to the application, which then presents or uses the result through the appropriate feature.

This structure allows AI functionality to be modified or improved without requiring major changes to the basic social networking modules.

5.3.7 Notification and Communication Services

Communication-related components connect different user activities with messaging and notification functions.

The messaging component manages communication between users, while the notification component provides updates related to relevant application events. These services interact with the database so that conversations and notification records can be stored and retrieved when required.

For example, a supported social interaction may result in a notification being created for the affected user. Similarly, a new message is stored and made available within the appropriate conversation.

5.3.8 Administration Layer

The Administration Layer provides authorized users with functions for managing and monitoring the application. It is separated from the normal user interface so that administrative operations can be controlled independently.

Depending on the implemented functionality, administrators can manage relevant user information, content, and application activities. This provides a centralized mechanism for maintaining the system and supporting administrative operations.

5.3.9 Architectural Benefits

The architecture provides several benefits to the development of NILA SOCIAL MEDIA:

1. **Modularity:** Individual functions are organized into separate components.
2. **Maintainability:** Changes to one component can be made with less impact on unrelated components.
3. **Scalability:** Additional modules and services can be introduced as the system develops.
4. **Security:** Authentication and access-control operations can be handled separately from the presentation layer.
5. **Reusability:** Common application services can be used by multiple features.
6. **AI Integration:** AI-assisted components can be connected with the application without making them dependent on every core feature.
7. **Simplified Testing:** Individual modules can be tested separately before complete system integration.

Overall, the architecture provides a structured foundation for NILA SOCIAL MEDIA by connecting the user interface, application logic, database, media services, communication features, and AI-assisted components in an organized manner. It also provides flexibility for future improvements while keeping the core application manageable

5.4 Case Study

The case study of **NILA SOCIAL MEDIA** describes the working of the system through a typical user scenario. It shows how different modules interact with each other to provide a complete social networking experience.

A new user first creates an account by providing the required registration details. After successful registration, the user logs into the system using the registered credentials. The authentication module verifies the details and provides access to the user's account. The user can then create a profile, update profile information, and manage privacy settings.

After setting up the profile, the user can upload photos or videos and create posts, stories, or reels. The uploaded content is processed and stored with the corresponding user information. Other users can view the content according to the applicable privacy settings and interact through likes and comments.

Users can also follow other accounts. For public accounts, the follow action can be processed directly, while private accounts may require approval. Follow activities and other relevant interactions can generate notifications for the concerned users.

The messaging module allows users to communicate with each other through direct messages. The search and explore module helps users find accounts and content based on available search information. These activities are connected with the database so that user actions and application data can be stored and retrieved when required.

NILA SOCIAL MEDIA also includes selected AI-assisted functionality. Depending on the implemented feature, AI can assist with tasks such as content assistance, recommendation-related processing, and image or content analysis. The AI component works together with the main application modules and provides additional support without affecting the basic social networking functions.

Overall, this case study demonstrates how authentication, profiles, content management, social interactions, privacy, messaging, notifications, search, database management, and AI-assisted features work together within NILA SOCIAL MEDIA. The case study also shows the practical use of the modular architecture discussed in the previous section.

CHAPTER 6
IMPLEMENTATION DETAILS

6.1 Implementation Procedures

The implementation of NILA SOCIAL MEDIA was carried out in a systematic manner by developing and integrating the major modules of the application. The development process focused on implementing the core social networking features first and then integrating supporting and AI-assisted functionalities.

The first step was to set up the development environment and create the basic project structure. The required programming tools, libraries, database, and development environment were configured before starting the implementation.

The authentication and user management modules were implemented first. These modules handle user registration, login, logout, profile management, and access to protected features. Database structures were created to store user and account-related information.

Next, the content management features were developed. Users can create posts and upload supported media. The implementation was extended to stories and reels based on the requirements of the application. The uploaded content is associated with the respective user and stored for later retrieval and display.

The social interaction modules were then implemented. Features such as likes, comments, following, and follow requests were connected with the user and content data. Privacy controls were also integrated to manage access to content and interactions for different types of accounts.

Communication-related features such as messaging and notifications were implemented after the basic social functions were completed. These modules allow users to communicate and receive updates about relevant activities within the application.

The search and explore functionality was added to help users find accounts and available content. Selected AI-assisted features were then integrated into the application to support tasks such as content assistance, recommendation-related processing, and image or content analysis, depending on the implemented functionality.

Finally, the different modules were integrated and tested together. Functional testing was performed to identify errors in authentication, content handling, social interactions, privacy, communication, and AI-assisted operations. Necessary corrections were made to improve the overall reliability and usability of the system.

6.2 Testing Methods and Results

Testing was carried out to verify that the major functions of **NILA SOCIAL MEDIA** work correctly and that the different modules interact properly. The testing process focused on identifying functional errors, checking user inputs, and ensuring that the expected output was produced.

6.2.1 Testing Methods

The following testing methods were used:

- **Unit Testing:** Individual functions and modules such as registration, login, posts, likes, comments, and notifications were checked separately.
- **Integration Testing:** Related modules were tested together to verify proper communication between authentication, database, social interaction, messaging, and other components.
- **Functional Testing:** Each major feature was tested using different inputs to confirm that it performs according to the specified requirements.
- **Validation Testing:** User inputs such as registration details, login credentials, comments, and uploaded media were checked for valid and invalid cases.
- **User Interface Testing:** The main screens and navigation options were checked to ensure that users can access and use the available features properly.

6.2.2 Testing Results

The testing process showed that the major implemented functions of NILA SOCIAL MEDIA performed according to their intended functionality. User registration and login were checked successfully, and authenticated users were able to access the appropriate features.

Post creation, media handling, likes, comments, follow operations, messaging, notifications, search, and privacy-related functions were tested individually and in

combination. The AI-assisted functions were also checked to verify that the input was processed and the corresponding output was returned.

During testing, errors identified in individual modules were corrected and the affected functions were tested again. The final testing confirmed that the integrated modules were working together as expected and that the system was suitable for demonstration and further development.

Testing Results

Module / Feature	**Test Performed**	**Result**
User Authentication	Valid and invalid login details	Passed
User Registration	Valid registration details and duplicate account check	Passed
Profile Management	Create, update, and view user profile	Passed
Post Management	Create and display photo/video posts	Passed
Likes and Comments	Add and remove likes; add comments	Passed
Follow Management	Follow, unfollow, and follow request operations	Passed
Stories	Create and view temporary stories	Passed
Reels	Upload and view short video content	Passed
Messaging	Send and receive messages between users	Passed
Notifications	Generate and display activity notifications	Passed
Search and Explore	Search for users and available content	Passed
Privacy Control	Test public and private account access	Passed
AI-Assisted Features	Test implemented AI-based content processing	Passed
Admin Module	Manage users and application content	Passed
Media Upload	Upload and retrieve supported media files	Passed

CHAPTER 7
ADVANTAGES AND DISADVANTAGES

7.1 Advantages

NILA SOCIAL MEDIA provides several advantages by bringing major social networking activities into a single web-based platform. It provides users with a common environment for content sharing, communication, interaction, privacy management, and selected AI-assisted functions.

- **Unified Social Networking Platform:** Profiles, posts, stories, reels, likes, comments, follows, messaging, notifications, and search are handled within the same platform.
- **Easy Content Sharing:** Users can create and share supported images and videos through posts, stories, and reels without using a separate content-sharing platform.
- **Interactive User Experience:** Likes, comments, follows, and other social interactions allow users to communicate and engage with shared content.
- **AI-Assisted Functionality:** Selected AI-based features provide support for content assistance, recommendation-related processing, and image or content analysis.
- **Privacy Management:** Users can manage account privacy and content visibility. Private account functionality can restrict access until the required follow request is approved.
- **Integrated Communication:** Messaging and notifications allow users to communicate directly and receive updates about relevant activities within the application.
- **Search and Content Discovery:** Search and explore features help users find accounts and available content using supported search information.
- **Centralised Data Management:** User accounts, profiles, posts, interactions, messages, notifications, and other application information can be maintained through the system database.
- **Modular Development:** The application is divided into separate modules, making individual features easier to develop, test, maintain, and modify.

- **Future Expandability:** The system provides a foundation for adding advanced AI models, improved recommendation techniques, enhanced security, and additional social networking features.

Overall, these advantages make **NILA SOCIAL MEDIA** a suitable academic platform for demonstrating the development of a modern social networking application. By combining social interaction, multimedia sharing, communication, privacy controls, and AI-assisted functionality, the system provides an organised environment for users while also offering opportunities for future development.

7.2 Disadvantages

Although **NILA SOCIAL MEDIA** provides several useful features, the system also has certain limitations. These limitations are mainly related to scalability, storage, AI processing, security, and the requirements of maintaining a multi-feature web application.

- **Limited Scalability:** The current system is mainly designed for academic development and demonstration. Supporting a large number of users simultaneously would require additional server and database optimization.
- **Internet Dependency:** Since the application is web-based, a stable internet connection is required for activities such as media uploads, messaging, notifications, and accessing online content.
- **Limited AI Capability:** The implemented AI-assisted functions are designed for selected tasks. More advanced AI models and larger datasets may be required to support complex real-world applications.
- **High Media Storage Requirement:** Images and videos can require considerable storage space. As the number of users and uploaded media increases, additional storage management would be necessary.
- **Basic Recommendation Processing:** Recommendation quality may be limited when only a small amount of user interaction data is available. More user activity and improved recommendation techniques can provide better personalization.
- **Security Maintenance:** Authentication and privacy controls provide basic protection, but a large-scale social networking platform would require continuous security testing, monitoring, and additional protection mechanisms.

- **Maintenance Complexity:** Since the system contains several interconnected modules, changes or updates to one feature may require testing of related modules to maintain compatibility.
- **Limited Real-Time Performance:** Some features may require further optimization to provide highly responsive real-time communication and notifications when the number of users increases.
- **Device and Browser Compatibility:** The web application may require additional optimization to provide a consistent experience across different browsers, screen sizes, and mobile devices.
- **Future Development Requirements:** Advanced features such as stronger AI models, large-scale deployment, improved recommendation systems, and enhanced security would require additional development, testing, and resources.

Overall, these limitations indicate areas where **NILA SOCIAL MEDIA** can be improved in future versions. The current system provides a functional academic foundation while leaving opportunities for further enhancement in scalability, AI capabilities, security, performance, and usability.

CHAPTER 8

RESULT AND DISCUSSION

8.1 Analysis

The analysis of **NILA SOCIAL MEDIA** was carried out based on the implementation and testing of its major modules. The results indicate that the developed system can support the main activities of a social networking platform within a single web-based environment. The core functions, including user authentication, profile management, content sharing, social interaction, messaging, notifications, search, privacy management, and selected AI-assisted features, were tested using suitable sample data and different input conditions.

The testing showed that users could register, log in, and access the system according to their authentication status. Profile management and content-sharing features allowed users to create and manage their information and share supported media through posts, stories, and reels. Social interaction features such as likes, comments, follows, and follow requests also worked according to the implemented functionality.

The analysis also showed that privacy controls could be applied to user accounts and content. Public and private account conditions were tested to verify that access was handled according to the defined settings. The messaging and notification modules supported communication between users and provided updates for relevant activities.

Another important observation was the coordination between different modules during normal user activities. For example, creating a post involves authentication, media handling, database storage, and content display. Similarly, follow requests and social interactions are connected with privacy and notification functions. This integration helped maintain user and application information in a structured manner.

The AI-assisted functionality was also examined as part of the system analysis. The implemented AI-supported features were tested with suitable inputs to verify that the required processing was performed and the resulting information was returned to the relevant application module. The analysis also indicates that the effectiveness of AI-based functions can be improved further by using larger datasets and more advanced models.

Overall, the results demonstrate that the major functional requirements identified for **NILA SOCIAL MEDIA** were addressed by the implemented system. At the same time, the

analysis highlights practical limitations such as dependence on internet connectivity, media storage requirements, available computing resources, and the need for further optimization for large-scale usage.

The key observations obtained from the analysis of the system include:

- **Successful User Authentication:** Users were able to register, log in, and access the available functions according to their authentication status.
- **Effective Profile Management:** Users could create and manage their profiles and associated account information.
- **Integrated Content Sharing:** Posts, stories, and reels could be created and displayed through the same platform.
- **Successful Social Interaction:** Users could perform supported activities such as likes, comments, follows, and follow requests.
- **Privacy Management:** Public and private account conditions were handled according to the implemented privacy rules.
- **Effective Communication:** Messaging and notifications supported communication and activity updates between users.
- **Search and Content Discovery:** Users could search for supported accounts and content using the available search functionality.
- **AI-Assisted Processing:** The implemented AI-supported functions processed suitable inputs and provided results to the corresponding application modules.
- **Module Integration:** The major modules worked together to support the overall workflow from user authentication and content sharing to interaction and communication.
- **Identified Limitations:** The analysis identified practical dependencies such as internet connectivity, media storage, computing resources, and the need for future scalability improvements.

Overall, the observations indicate that **NILA SOCIAL MEDIA** achieved the major functional requirements defined for the proposed social networking platform while providing a foundation for future improvements in AI capabilities, security, scalability, and user experience.

CHAPTER 9

CONCLUSION

9.1 Summary

NILA SOCIAL MEDIA is a web-based social networking application developed to provide users with a common platform for communication, multimedia sharing, social interaction, and content discovery. The system combines important social networking functions such as user registration and authentication, profile management, posts, stories, reels, likes, comments, follow management, messaging, notifications, search, privacy controls, and administration.

The project was developed using a modular approach, where different functionalities are organized into separate modules. This structure makes the system easier to develop, test, maintain, and improve. Authentication and access control help protect user accounts, while privacy features provide users with control over their social connections and shared content. Media management supports the handling of image and video content within the application.

Selected AI-assisted functionality has also been included to support tasks such as content assistance, recommendation-related processing, and image or content analysis, depending on the implemented modules. These features demonstrate how AI techniques can be integrated with a conventional web-based social networking system.

The major modules were tested to verify important operations such as registration, login, profile management, content sharing, social interactions, messaging, notifications, search, privacy management, and AI-assisted processing. The testing process helped identify functional issues and improve the overall integration of the system.

Overall, NILA SOCIAL MEDIA provides a practical academic implementation of a modern social networking platform. The project establishes a foundation for future improvements such as more advanced AI models, improved recommendation techniques, stronger security mechanisms, better scalability, enhanced real-time communication, and mobile application support.

9.2 Limitations

NILA SOCIAL MEDIA includes the major features required for a social networking application, but some limitations remain due to the scope and resources of the project. These limitations mainly concern scalability, AI functionality, media storage, security, and system maintenance.

- Limited Scalability: The application is mainly developed for academic purposes and testing. Supporting a very large number of users would require additional server, database, and performance optimization.
- Limited AI Functionality: The AI-assisted features are designed for selected tasks. More advanced AI functions would require larger datasets, better models, and higher computational resources.
- Media Storage: The application supports image and video sharing, which can require considerable storage space as the amount of user content increases.
- Basic Recommendation Processing: Recommendation-related features may be limited by the available user interaction data. More personalized recommendations would require larger datasets and advanced techniques.
- Internet Dependency: Features such as media sharing, messaging, notifications, and content access depend on a stable internet connection.
- Security Requirements: The application provides authentication and privacy controls, but a real-world large-scale platform would require continuous security testing and additional protection mechanisms.
- Maintenance and Compatibility: Since the system contains several connected modules, regular updates and testing are required. Further optimization may also be needed for different devices, browsers, and screen sizes.

Overall, NILA SOCIAL MEDIA provides a practical academic implementation of a modern social networking platform. The project establishes a foundation for future improvements such as more advanced AI models, improved recommendation techniques, stronger security mechanisms, better scalability, enhanced real-time communication, and mobile application support.

9.3 Future Scope

NILA SOCIAL MEDIA provides a basic foundation for developing a more advanced and intelligent social networking platform. The system can be extended in several ways to improve its performance, security, intelligence, and user experience.

- **Advanced AI Features:** More advanced AI models can be integrated for automatic content generation, image understanding, content moderation, and personalized user assistance.
- **Improved Recommendation System:** The recommendation module can be enhanced using larger amounts of user interaction data to provide more personalized posts, reels, and user suggestions.
- **Enhanced Security:** Stronger authentication, improved privacy controls, encryption, and advanced security mechanisms can be added to protect user accounts and data.
- **Better Scalability:** Cloud-based services, optimized databases, and improved server architecture can be introduced to support a larger number of users and multimedia files.
- **Real-Time Communication:** Advanced real-time messaging, online status, typing indicators, voice calls, and video calls can be included to improve communication.
- **Mobile Application:** A dedicated Android and iOS application can be developed so that users can access NILA SOCIAL MEDIA conveniently through mobile devices.
- **Multilingual Support:** Support for multiple languages can be added to make the platform accessible to users from different regions.
- **Advanced Analytics:** User and creator analytics can be introduced to provide useful information about content engagement, reach, and interaction patterns.
- **Improved Media Features:** Additional features such as advanced video editing, filters, effects, and improved media processing can be added to enhance content creation.

Overall, these future enhancements can transform NILA SOCIAL MEDIA into a more scalable, secure, intelligent, and user-friendly social networking platform.

9.4 Applications

NILA SOCIAL MEDIA can be used in different areas where communication, content sharing, and social interaction are required. It provides users with a common platform to create profiles, share multimedia content, communicate with others, and participate in online activities. The system can also be adapted for different types of users and communities based on their specific requirements. Its modular structure makes it possible to extend the existing features and develop additional functions for educational, promotional, community-based, and other social networking purposes.

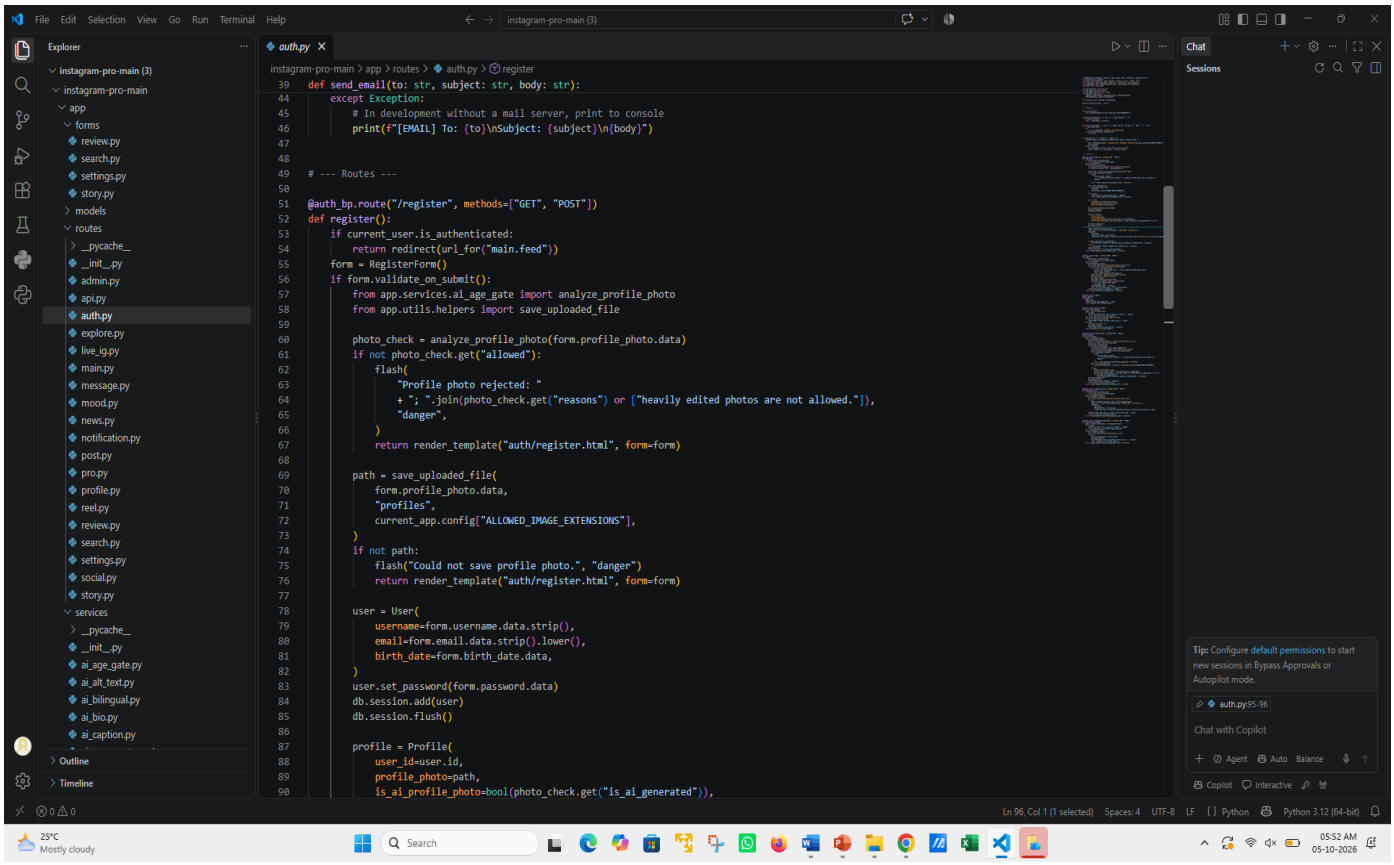
- **Social Networking:** Users can create profiles, share posts, stories, and reels, and interact through likes, comments, and follows.
- **Personal Communication:** Messaging and notifications allow users to communicate and stay updated about different activities.
- **Content Sharing:** Users can share images, videos, captions, and other digital content with their followers and other users.
- **Community Building:** The platform can support communities where users with common interests can share ideas and interact with each other.
- **Educational Use:** It can be adapted for students and teachers to share study materials, announcements, project updates, and other academic content.
- **Promotional Activities:** Businesses, creators, and organizations can use similar features to promote their products, services, events, or content.
- **AI-Assisted Applications:** The platform provides a foundation for using AI in content assistance, recommendations, and image or content analysis.
- **Academic and Research Use:** The system can be used for learning and research in web development, database management, social networking, and AI integration.

Overall, NILA SOCIAL MEDIA provides a flexible platform for social networking, communication, content sharing, education, promotion, and AI-assisted applications. Its integrated features allow users to communicate, share content, and interact easily. The modular design also provides scope for future improvements and adaptation to different user requirements.

CHAPTER 10

APPENDICES

10.1 Sample Code



```python
def send_email(to: str, subject: str, body: str):
    except Exception:
        # In development without a mail server, print to console
        print(f"[EMAIL] To: {to}\nSubject: {subject}\n{body}")


# --- Routes ---

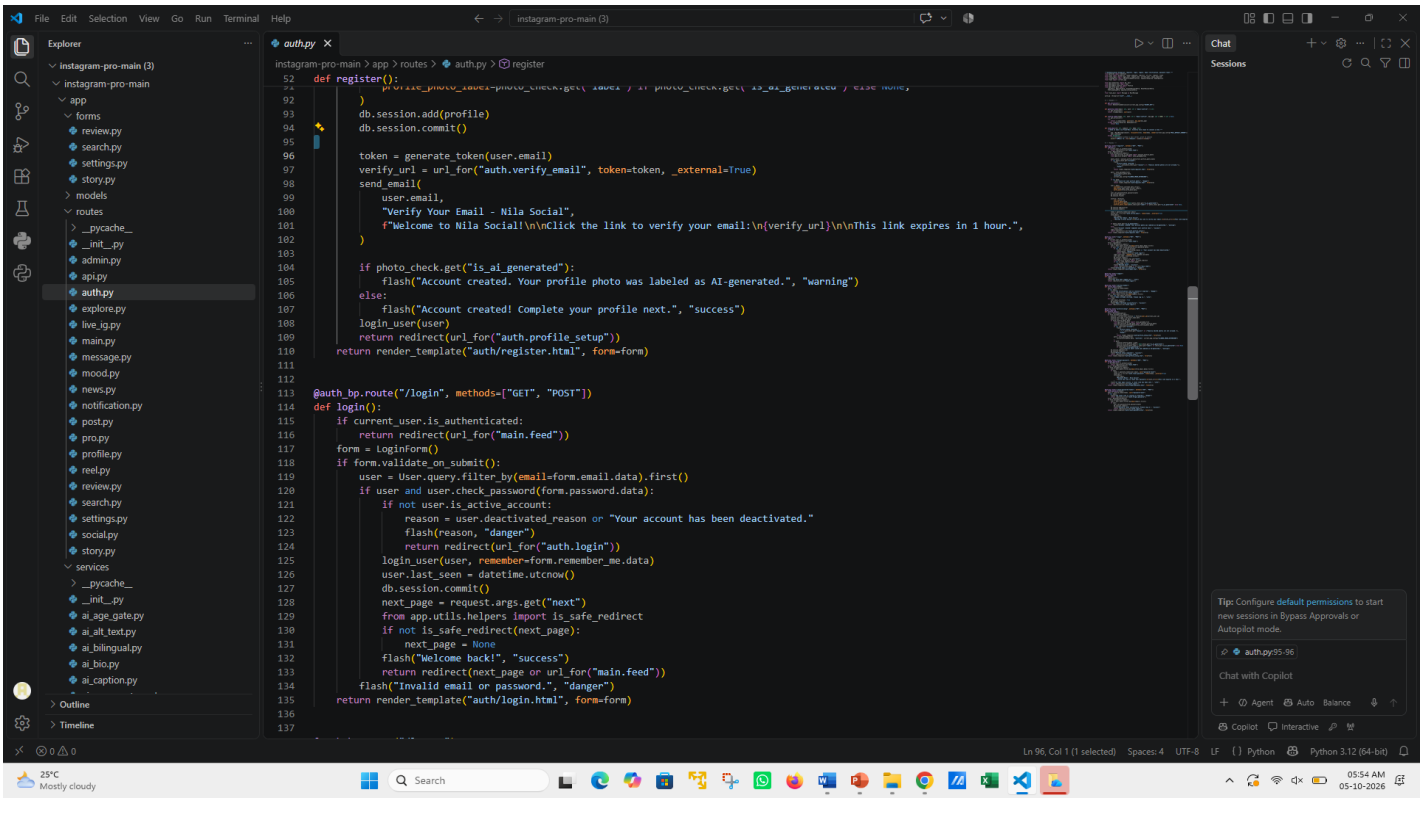
@auth_bp.route("/register", methods=["GET", "POST"])
def register():
    if current_user.is_authenticated:
        return redirect(url_for("main.feed"))
    form = RegisterForm()
    if form.validate_on_submit():
        from app.services.ai_age_gate import analyze_profile_photo
        from app.utils.helpers import save_uploaded_file

        photo_check = analyze_profile_photo(form.profile_photo.data)
        if not photo_check.get("allowed"):
            flash(
                "Profile photo rejected: "
                + "; ".join(photo_check.get("reasons") or ["heavily edited photos are not allowed."]),
                "danger",
            )
            return render_template("auth/register.html", form=form)

        path = save_uploaded_file(
            form.profile_photo.data,
            "profiles",
            current_app.config["ALLOWED_IMAGE_EXTENSIONS"],
        )
        if not path:
            flash("Could not save profile photo.", "danger")
            return render_template("auth/register.html", form=form)

        user = User(
            username=form.username.data.strip(),
            email=form.email.data.strip().lower(),
            birth_date=form.birth_date.data,
        )
        user.set_password(form.password.data)
        db.session.add(user)
        db.session.flush()

        profile = Profile(
            user_id=user.id,
            profile_photo=path,
            is_ai_profile_photo=bool(photo_check.get("is_ai_generated")),
```

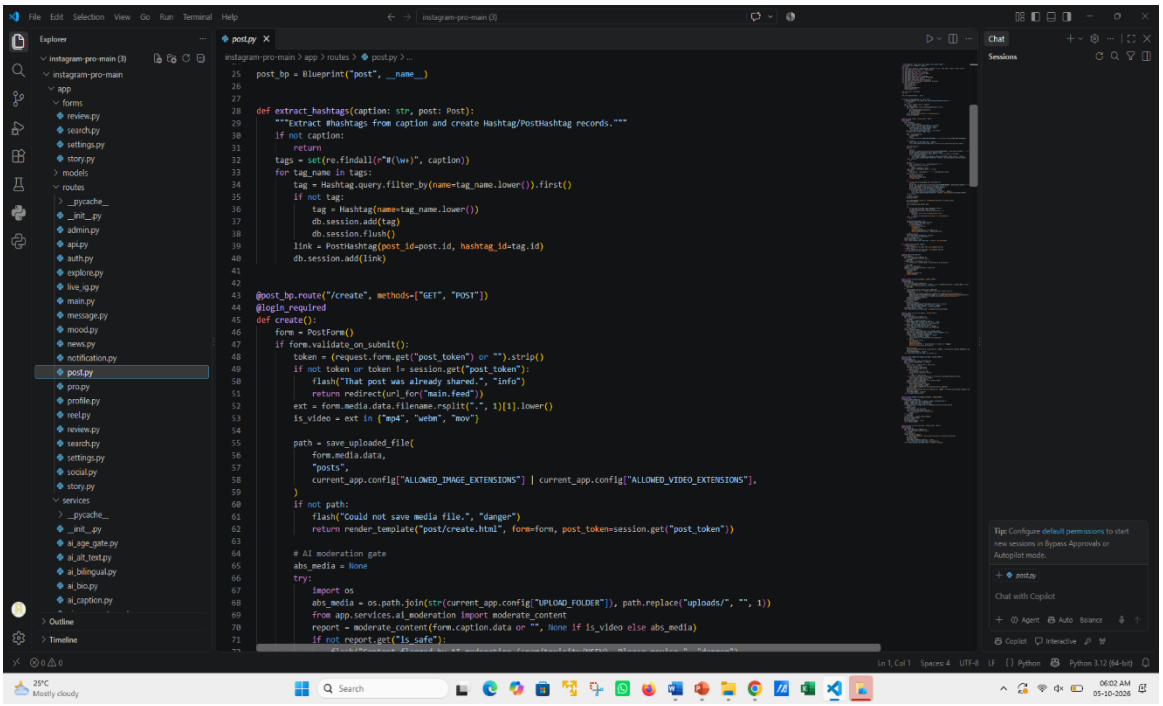


```python
        )
        db.session.add(profile)
        db.session.commit()

        token = generate_token(user.email)
        verify_url = url_for("auth.verify_email", token=token, _external=True)
        send_email(
            user.email,
            "Verify Your Email - Nila Social",
            f"Welcome to Nila Social!\n\nClick the link to verify your email:\n{verify_url}\n\nThis link expires in 1 hour.",
        )

        if photo_check.get("is_ai_generated"):
            flash("Account created. Your profile photo was labeled as AI-generated.", "warning")
        else:
            flash("Account created! Complete your profile next.", "success")
        login_user(user)
        return redirect(url_for("auth.profile_setup"))
    return render_template("auth/register.html", form=form)


@auth_bp.route("/login", methods=["GET", "POST"])
def login():
    if current_user.is_authenticated:
        return redirect(url_for("main.feed"))
    form = LoginForm()
    if form.validate_on_submit():
        user = User.query.filter_by(email=form.email.data).first()
        if user and user.check_password(form.password.data):
            if not user.is_active_account:
                reason = user.deactivated_reason or "Your account has been deactivated."
                flash(reason, "danger")
                return redirect(url_for("auth.login"))
            login_user(user, remember=form.remember_me.data)
            user.last_seen = datetime.utcnow()
            db.session.commit()
            next_page = request.args.get("next")
            from app.utils.helpers import is_safe_redirect
            if not is_safe_redirect(next_page):
                next_page = None
            flash("Welcome back!", "success")
            return redirect(next_page or url_for("main.feed"))
        flash("Invalid email or password.", "danger")
    return render_template("auth/login.html", form=form)
```



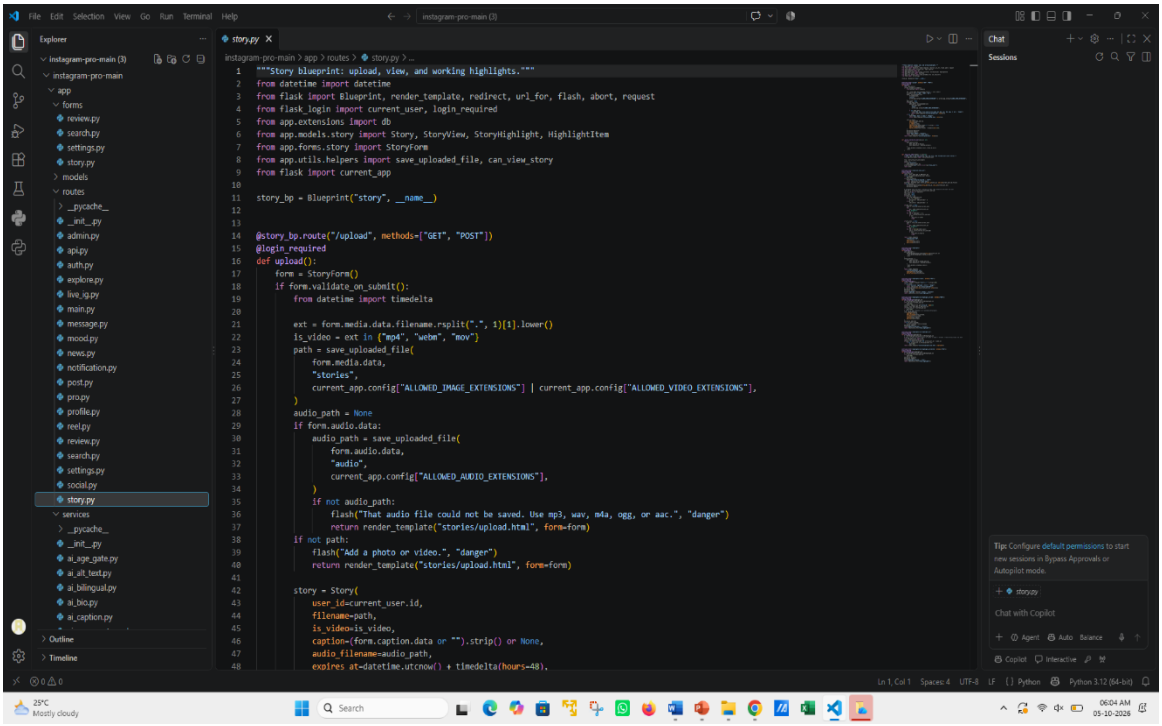
```python
post_bp = Blueprint("post", __name__)


def extract_hashtags(caption: str, post: Post):
    """Extract #hashtags from caption and create Hashtag/PostHashtag records."""
    if not caption:
        return
    tags = set(re.findall(r"#(\w+)", caption))
    for tag_name in tags:
        tag = Hashtag.query.filter_by(name=tag_name.lower()).first()
        if not tag:
            tag = Hashtag(name=tag_name.lower())
            db.session.add(tag)
            db.session.flush()
        link = PostHashtag(post_id=post.id, hashtag_id=tag.id)
        db.session.add(link)


@post_bp.route("/create", methods=["GET", "POST"])
@login_required
def create():
    form = PostForm()
    if form.validate_on_submit():
        token = (request.form.get("post_token") or "").strip()
        if not token or token != session.get("post_token"):
            flash("That post was already shared.", "info")
            return redirect(url_for("main.feed"))
        ext = form.media.data.filename.rsplit(".", 1)[1].lower()
        is_video = ext in {"mp4", "webm", "mov"}

        path = save_uploaded_file(
            form.media.data,
            "posts",
            current_app.config["ALLOWED_IMAGE_EXTENSIONS"] | current_app.config["ALLOWED_VIDEO_EXTENSIONS"],
        )
        if not path:
            flash("Could not save media file.", "danger")
            return render_template("post/create.html", form=form, post_token=session.get("post_token"))

        # AI moderation gate
        abs_media = None
        try:
            import os
            abs_media = os.path.join(str(current_app.config["UPLOAD_FOLDER"]), path.replace("uploads/", "", 1))
            from app.services.ai_moderation import moderate_content
            report = moderate_content(form.caption.data or "", None if is_video else abs_media)
            if not report.get("is_safe"):
```



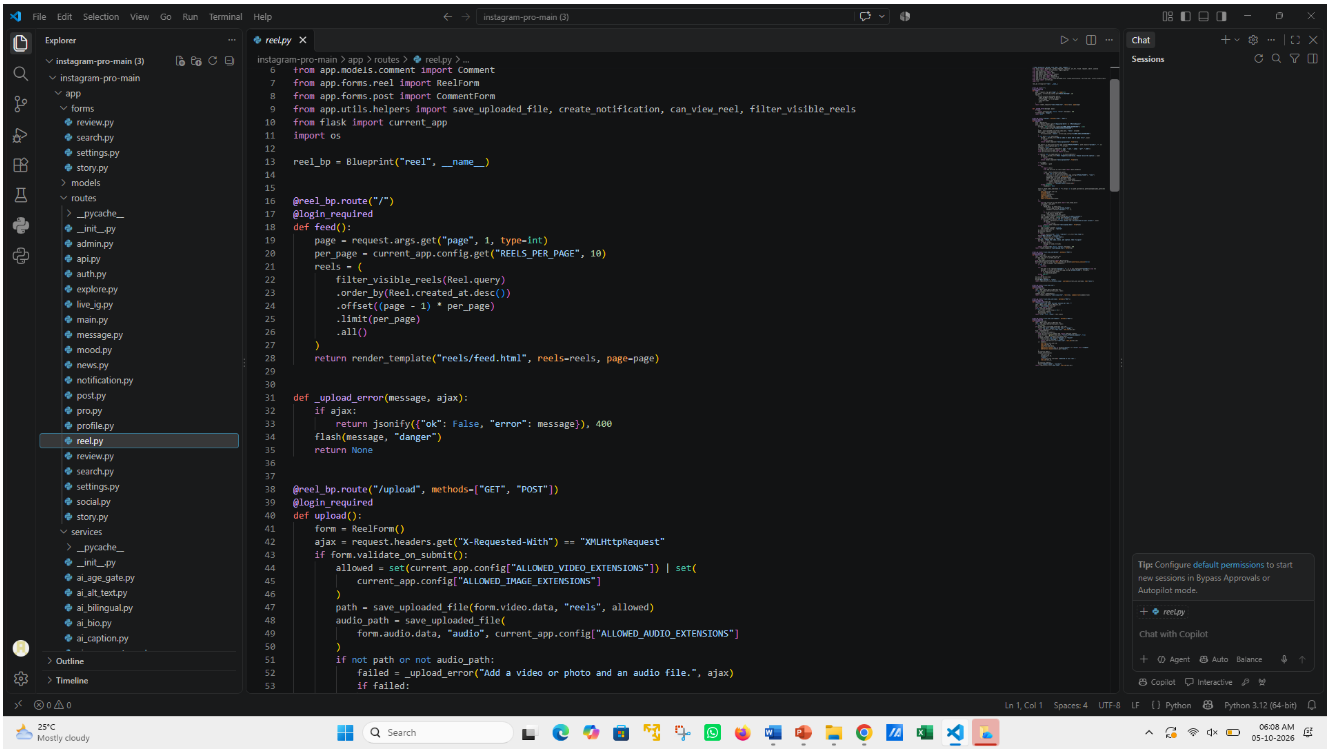
```python
"""Story blueprint: upload, view, and working highlights."""
from datetime import datetime
from flask import Blueprint, render_template, redirect, url_for, flash, abort, request
from flask_login import current_user, login_required
from app.extensions import db
from app.models.story import Story, StoryView, StoryHighlight, HighlightItem
from app.forms.story import StoryForm
from app.utils.helpers import save_uploaded_file, can_view_story
from flask import current_app

story_bp = Blueprint("story", __name__)


@story_bp.route("/upload", methods=["GET", "POST"])
@login_required
def upload():
    form = StoryForm()
    if form.validate_on_submit():
        from datetime import timedelta

        ext = form.media.data.filename.rsplit(".", 1)[1].lower()
        is_video = ext in {"mp4", "webm", "mov"}
        path = save_uploaded_file(
            form.media.data,
            "stories",
            current_app.config["ALLOWED_IMAGE_EXTENSIONS"] | current_app.config["ALLOWED_VIDEO_EXTENSIONS"],
        )
        audio_path = None
        if form.audio.data:
            audio_path = save_uploaded_file(
                form.audio.data,
                "audio",
                current_app.config["ALLOWED_AUDIO_EXTENSIONS"],
            )
            if not audio_path:
                flash("That audio file could not be saved. Use mp3, wav, m4a, ogg, or aac.", "danger")
                return render_template("stories/upload.html", form=form)
        if not path:
            flash("Add a photo or video.", "danger")
            return render_template("stories/upload.html", form=form)

        story = Story(
            user_id=current_user.id,
            filename=path,
            is_video=is_video,
            caption=(form.caption.data or "").strip() or None,
            audio_filename=audio_path,
            expires_at=datetime.utcnow() + timedelta(hours=48),
```



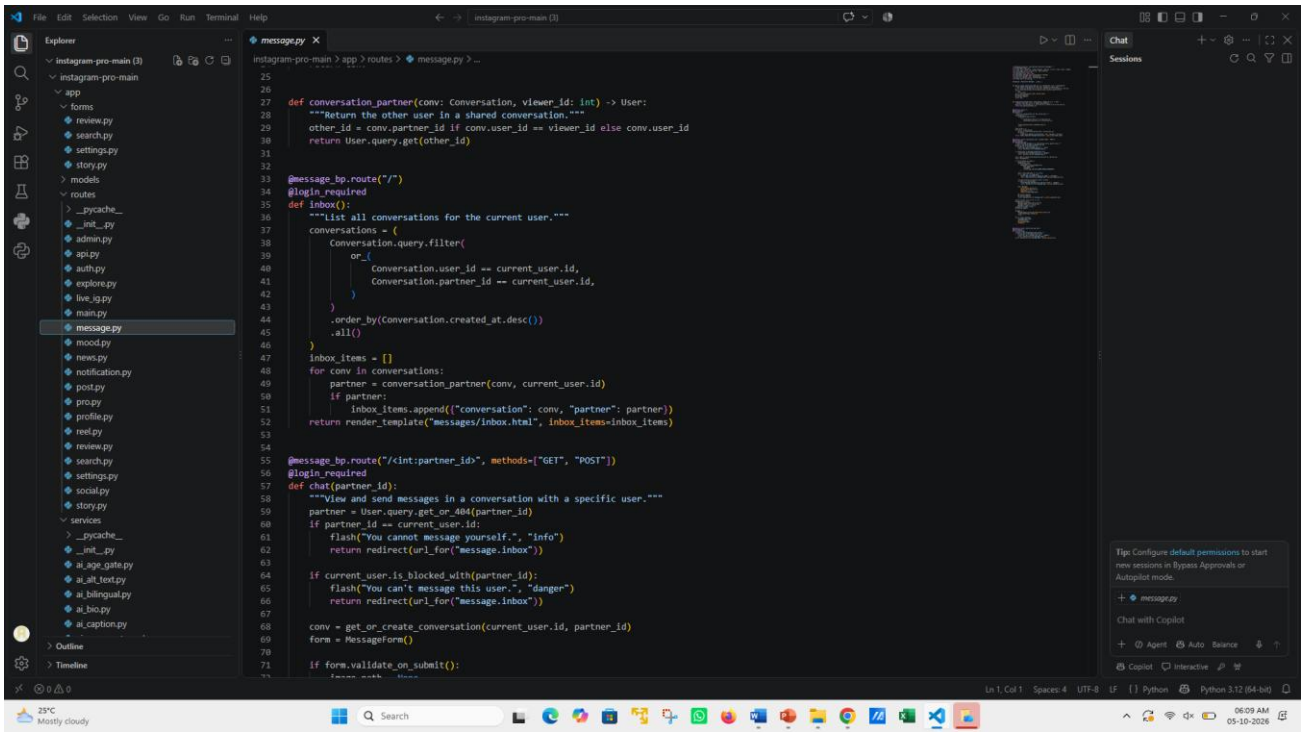
```python
from app.models.comment import Comment
from app.forms.reel import ReelForm
from app.forms.post import CommentForm
from app.utils.helpers import save_uploaded_file, create_notification, can_view_reel, filter_visible_reels
from flask import current_app
import os

reel_bp = Blueprint("reel", __name__)


@reel_bp.route("/")
@login_required
def feed():
    page = request.args.get("page", 1, type=int)
    per_page = current_app.config.get("REELS_PER_PAGE", 10)
    reels = (
        filter_visible_reels(Reel.query)
        .order_by(Reel.created_at.desc())
        .offset((page - 1) * per_page)
        .limit(per_page)
        .all()
    )
    return render_template("reels/feed.html", reels=reels, page=page)


def _upload_error(message, ajax):
    if ajax:
        return jsonify({"ok": False, "error": message}), 400
    flash(message, "danger")
    return None


@reel_bp.route("/upload", methods=["GET", "POST"])
@login_required
def upload():
    form = ReelForm()
    ajax = request.headers.get("X-Requested-With") == "XMLHttpRequest"
    if form.validate_on_submit():
        allowed = set(current_app.config["ALLOWED_VIDEO_EXTENSIONS"]) | set(
            current_app.config["ALLOWED_IMAGE_EXTENSIONS"]
        )
        path = save_uploaded_file(form.video.data, "reels", allowed)
        audio_path = save_uploaded_file(
            form.audio.data, "audio", current_app.config["ALLOWED_AUDIO_EXTENSIONS"]
        )
        if not path or not audio_path:
            failed = _upload_error("Add a video or photo and an audio file.", ajax)
            if failed:
```



```python
def conversation_partner(conv: Conversation, viewer_id: int) -> User:
    """Return the other user in a shared conversation."""
    other_id = conv.partner_id if conv.user_id == viewer_id else conv.user_id
    return User.query.get(other_id)


@message_bp.route("/")
@login_required
def inbox():
    """List all conversations for the current user."""
    conversations = (
        Conversation.query.filter(
            or_(
                Conversation.user_id == current_user.id,
                Conversation.partner_id == current_user.id,
            )
        )
        .order_by(Conversation.created_at.desc())
        .all()
    )
    inbox_items = []
    for conv in conversations:
        partner = conversation_partner(conv, current_user.id)
        if partner:
            inbox_items.append({"conversation": conv, "partner": partner})
    return render_template("messages/inbox.html", inbox_items=inbox_items)


@message_bp.route("/<int:partner_id>", methods=["GET", "POST"])
@login_required
def chat(partner_id):
    """View and send messages in a conversation with a specific user."""
    partner = User.query.get_or_404(partner_id)
    if partner_id == current_user.id:
        flash("You cannot message yourself.", "info")
        return redirect(url_for("message.inbox"))

    if current_user.is_blocked_with(partner_id):
        flash("You can't message this user.", "danger")
        return redirect(url_for("message.inbox"))

    conv = get_or_create_conversation(current_user.id, partner_id)
    form = MessageForm()

    if form.validate_on_submit():
```

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Vaishnavi Gireesh

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